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# PCCST 501 — COMPUTER NETWORKS

## MODULE 4

*SNMP & ASN.1 | Physical Layer: Data & Signals, Digital & Analog Transmission, Bandwidth Utilization, Transmission Media*

Semester S5 — Computer Networks (PCCST501)

*Common to CS/CD/CM/CR/CA/AD/AI/CB/CN/CU/CI*

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# PART A — NETWORK MANAGEMENT: SNMP

## 1. Introduction to Network Management

As networks grow larger and more heterogeneous — spanning multiple vendors, technologies, and physical media — administrators need a standardized way to monitor device health, diagnose faults, and configure remote equipment without physically visiting each device. This is the job of network management protocols, of which SNMP is the most widely deployed.

Network management broadly involves five functional areas, often remembered by the acronym FCAPS:

| Area | What it covers |
|---|---|
| Fault Management | Detecting, logging, and resolving network faults (e.g., a link going down). |
| Configuration Management | Tracking and modifying device configurations. |
| Accounting Management | Tracking resource usage for billing or auditing. |
| Performance Management | Monitoring throughput, latency, error rates, etc. |
| Security Management | Controlling access to network resources and management data. |

SNMP primarily supports fault and performance management, while working alongside SMI and MIB (explained later) to standardize how management data is described and organized.

## 2. SNMP — Simple Network Management Protocol

SNMP is an application-layer protocol (it sits at the top of the TCP/IP protocol stack, typically running over UDP on ports 161 and 162) used to manage and monitor network devices such as routers, switches, and servers.

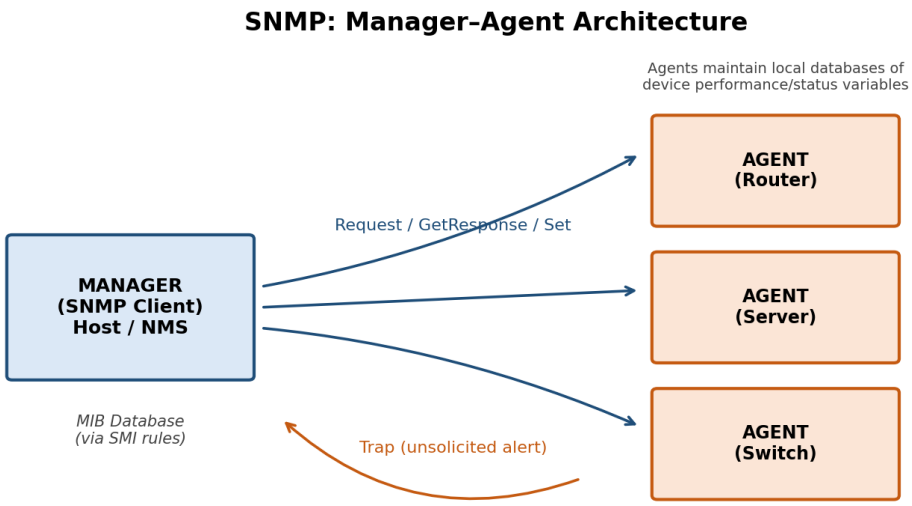
SNMP is deliberately kept 'simple' — its command set is small, and it treats every managed device the same way: as a collection of named variables that can be read or, in some cases, written.

### 2.1 Why SNMP is needed

Because SNMP operates at the application level and defines its own abstract way of naming and describing data (via SMI/ASN.1), it is independent of:

- The device manufacturer (Cisco, Juniper, HP, etc. can all implement the same SNMP standard)
- The underlying physical network technology (Ethernet, Wi-Fi, fiber, etc.)
- Whether the device sits on a LAN or a WAN

This separation of the management protocol from the underlying hardware is what makes SNMP a universal, vendor-neutral standard for network management.




## 2.2 Manager and Agent

SNMP works using a client–server model with two main components:

**Manager (SNMP client):** Usually a host or Network Management Station (NMS) that runs SNMP client software. The manager actively monitors devices, requests data from agents, can modify values in an agent's database, and can trigger agents to perform actions (e.g. reboot).

**Agent (SNMP server):** A router, switch, server, or host running SNMP server software. The agent passively maintains a local database of performance/status information, responds to manager requests, and can proactively notify (trap) the manager of abnormal events.

| Direction | Interaction |
|---|---|
| Manager → Agent | Monitors and controls (sends Get, GetNext, Set requests) |
| Agent → Manager | Provides information and warnings (sends GetResponse, Trap) |

## 2.3 How the Manager and Agent Work Together

Each agent continuously updates a local database with statistics about the device it runs on — for example, a router might track the number of packets received, packets forwarded, and error counts. The manager periodically polls (queries) these values and compares them to detect abnormal patterns.

**Worked Example — Detecting Congestion:** Suppose a router agent reports Packets received = 10,000 and Packets forwarded = 6,000. The manager can calculate that roughly 4,000 packets (40%) were not forwarded, which could indicate congestion, buffer overflow, or a misconfigured routing table — prompting the administrator to investigate further.

### 2.4 The Manager Can Control the Agent

Beyond simply reading values, SNMP's Set operation allows a manager to write new values into an agent's MIB variables, which can trigger real actions on the device.

**Worked Example — Remote Reboot:** A router may expose a writable 'reboot counter' object. Setting this counter to 0 via SNMP causes the router's firmware to detect the change and initiate a reboot — allowing administrators to restart remote equipment without physical access.

### 2.5 The Agent Can Warn the Manager — Traps

Polling every device periodically for every possible fault would be inefficient and slow to detect emergencies. To solve this, SNMP allows agents to proactively send unsolicited trap messages to the manager whenever they detect a serious or abnormal condition (e.g., interface down, high temperature, authentication failure), instead of waiting to be asked.

**Agent → Trap (unsolicited alert) → Manager**

### 2.6 The Three Basic Ideas of SNMP Management

All SNMP operations can be summarized into three fundamental interaction patterns:

1. Manager checks the Agent — the manager requests information (polling) to monitor behaviour and performance. Manager → Request → Agent
2. Manager controls the Agent — the manager changes a value in the agent's database to make it perform a task. Manager → Set value → Agent performs task
3. Agent warns the Manager — the agent detects an unusual condition and sends an unsolicited trap. Agent → Trap → Manager

### 2.7 SNMP Protocol Data Units (PDUs) —

Although the slides summarize SNMP at a conceptual level, exam questions frequently ask about the actual message types (PDUs) SNMP defines. These are summarized below for completeness:

| PDU / Message | Direction | Purpose |
|---|---|---|
| GetRequest | Manager → Agent | Retrieve the value of one or more MIB objects. |
| GetNextRequest | Manager → Agent | Retrieve the next object in the MIB tree (used to 'walk' a MIB). |

| PDU / Message | Direction | Purpose |
|---|---|---|
| GetBulkRequest | Manager → Agent | Efficiently retrieve large blocks of table data (SNMPv2/v3). |
| SetRequest | Manager → Agent | Modify the value of a MIB object. |
| GetResponse / Response | Agent → Manager | Reply to a Get/Set request, carrying requested values or an error. |
| Trap / TrapV2 / InformRequest | Agent → Manager | Unsolicited notification of an event or abnormal condition. |

## 3. SNMP Management Components: SNMP, SMI and MIB

Network management using SNMP actually rests on three cooperating components, each with a distinct role:

| Component | Full Form | Role |
|---|---|---|
| SNMP | Simple Network Management Protocol | Defines HOW the manager and agent communicate — the packet format, requests, and responses. |
| SMI | Structure of Management Information | Defines the RULES for naming and describing managed objects (data types, hierarchy). |
| MIB | Management Information Base | Defines WHICH objects are actually managed for a given device — the database of names. |

**Note:** An easy way to remember the three-layer relationship: SMI is the grammar, MIB is the dictionary (built using that grammar), and SNMP is the language used to exchange sentences built from that dictionary.

### 3.1 Role of SNMP

SNMP defines the communication mechanics between manager and agent:

- Defines the format of the packets exchanged between manager and agent.
- Carries object names together with their current values or status.
- Allows the manager to read (Get) and modify (Set) values.

- Helps interpret received information and generate statistics for the administrator.

### 3.2 Role of SMI

SMI does not define the specific objects to be managed — that is MIB's job. Instead, SMI defines the general rules for describing management objects, specifying:

- How objects are named (using a globally unique hierarchical identifier, e.g. an OID such as 1.3.6.1.2.1.1).
- How objects are organized hierarchically (a tree structure, similar to a file-system directory tree).
- What data types can be used to represent object values (e.g. INTEGER, OCTET STRING, Counter, Gauge).
- How data should be encoded/represented so that different devices, from different manufacturers, can understand it identically.

SMI itself relies on ASN.1 (discussed in Section 4) as the underlying notation for defining these data types.

### 3.3 Role of MIB

The MIB is the actual database of managed objects for a specific class of device. It is essentially a standardized, hierarchical list of variable names (like 'ifInOctets' for interface input byte count) that both the manager and agent agree to use, so that a request for a given object name always means the same thing everywhere.

- Lists the managed objects available on a device (e.g. sysUpTime, ifNumber, ifSpeed).
- Gives each object a unique name following SMI's naming rules (an Object Identifier, OID).
- Associates a specific data type with each object.
- Functions like a database schema — the structure that holds object names and their types, while SNMP is used to actually read/write the current values.

## 4. ASN.1 — Abstract Syntax Notation One

ASN.1 is a formal, platform-independent notation (a language) used to define the format and structure of data exchanged between different computer systems. SMI uses a restricted subset of ASN.1 to define MIB object types.

### 4.1 Why ASN.1 is Needed

When two systems exchange raw bits over a network, there is no inherent meaning attached to those bits — the receiver must somehow know how to interpret them. Suppose a sender wants to transmit 'a name and a number.' The receiving system needs answers to questions such as:

- Which bits represent the name, and which represent the number?
- What is the boundary between one field and the next?
- How are complex, composite data structures such as arrays or records organized and transmitted?

ASN.1 solves this by providing a standard, machine-independent notation for describing data structures, completely separate from how those structures are eventually encoded into bits for transmission (that encoding step is handled by a companion standard called BER — Basic Encoding Rules).

### 4.2 ASN.1 Symbols

| Symbol | Meaning | Symbol | Meaning |
|---|---|---|---|
| ::= | Defined as / assignment | .. | Range |
| \| | Or, alternative, option | { } | Start and end of a list |
| – | Negative sign | [ ] | Start and end of a tag |
| -- | The following is a comment | ( ) | Start and end of a subtype |

### 4.3 ASN.1 Data Types

ASN.1 provides a set of built-in ('primitive') data types:

| Type | Meaning |
|---|---|
| INTEGER | Whole numbers (positive, negative, or zero). |
| REAL | Real (floating-point) numbers. |
| BOOLEAN | TRUE / FALSE values. |
| ENUMERATED | A fixed, named list of choices (like an enum in C). |
| SEQUENCE | An ordered list/record of values (like a struct). |
| SET | An unordered collection of values. |
| VisibleString | A string of printable (visible) characters. |

### 4.4 Creating New Data Types

New, named types can be built from the built-in primitive types. By convention, type names always begin with a capital letter:

```
Age ::= INTEGER

MaritalStatus ::= ENUMERATED { single, married, widowed, divorced }

DayOfWeek ::= ENUMERATED { sun, mon, tue, wed, thu, fri, sat }
```

### 4.5 Creating Subtypes (Restricting a Range)

ASN.1 also allows an existing type's range of legal values to be restricted, producing a subtype:

```
NumberOfStudents ::= INTEGER (15..40)

Grade ::= REAL (1.0..5.0)

Weekend ::= DayOfWeek (sun | sat)
```

- NumberOfStudents can only take integer values from 15 to 40.
- Grade can only take real values from 1.0 to 5.0.
- Weekend can only be Sunday or Saturday (a restricted choice from DayOfWeek).

### 4.6 Variables

In ASN.1, a variable is declared with both a type and an assigned value. By convention, variable names always begin with a lowercase letter (this is the opposite convention to type names, and a common source of confusion/exam questions):

```
numberOfComputers INTEGER ::= 2

herAge INTEGER ::= 35

classSize NumberOfStudents ::= 22
```

### 4.7 Structured Data — SEQUENCE

SEQUENCE is used to build a structured, record-like data type, analogous to a struct in C/C++ — an ordered collection of named fields, each possibly of a different type.

```
StudentAccount ::= SEQUENCE { userName VisibleString, password VisibleString, accountNumber INTEGER }
```

A value of this composite type can then be assigned by providing each field:

```
johnNewton StudentAccount ::= { userName "JohnN", password "120007", accountNumber 25579 }
```

### 4.8 SEQUENCE OF — Collections of Similar Items

SEQUENCE OF is used when a collection of many similar items is needed, analogous to an array in C/C++. For instance, a router's forwarding table — which contains many rows of the same structure — can be represented as:

```
ForwardingTable ::= SEQUENCE OF Row
```

where each Row is itself a SEQUENCE containing several fields (e.g., destination address, next hop, interface, metric).

### 4.9 Summary: SNMP, SMI, MIB and ASN.1 Working Together

To manage a router's interface counters, for example: SMI/ASN.1 define the data type of the counter (e.g. an unsigned INTEGER, or 'Counter32'); the MIB assigns this object a standard name/OID (e.g. ifInOctets); and SNMP is the protocol actually used by the manager to send a GetRequest for that OID and receive the current value back from the agent.

# PART B — THE PHYSICAL LAYER

## 5. Introduction to the Physical Layer

The Physical Layer is Layer 1 of the OSI model. It is responsible for the actual, node-to-node transmission of raw bits over a physical transmission medium. It takes the stream of bits handed down from the Data-Link Layer and converts it into signals suitable for the chosen medium.

**Bits → Electromagnetic Signals → Transmission Medium**

The Physical Layer deals with several closely related concerns, each expanded in the sections that follow:

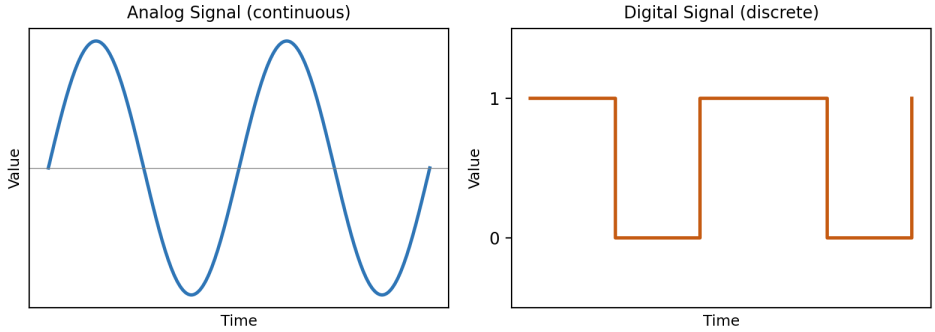
- Digital and analog signals — how information is represented physically.
- Digital and analog transmission — how data (digital or analog) is converted into a signal suitable for the channel.
- Data transmission speed and channel capacity — how fast data can reliably be sent.
- Multiplexing — how multiple signals can efficiently share one physical medium.
- Spread spectrum — how bandwidth is deliberately expanded for security and robustness.
- Transmission media — the guided (cabled) and unguided (wireless) paths signals travel through.

## 6. Data and Signals

Communication at the Physical Layer takes place node-to-node using electromagnetic signals travelling across a medium. Before signals can be studied, it is essential to distinguish between data (the information itself) and signals (the physical representation used to carry that data).

### 6.1 Analog and Digital Data

| Analog Data | Digital Data |
|---|---|
| Continuous in nature — can take on infinitely many values within a range. | Discrete in nature — takes on a finite, countable set of values. |
| Example: human voice, temperature readings, sound waves. | Example: text, numbers, and other computer-generated data, usually represented as sequences of 0s and 1s. |



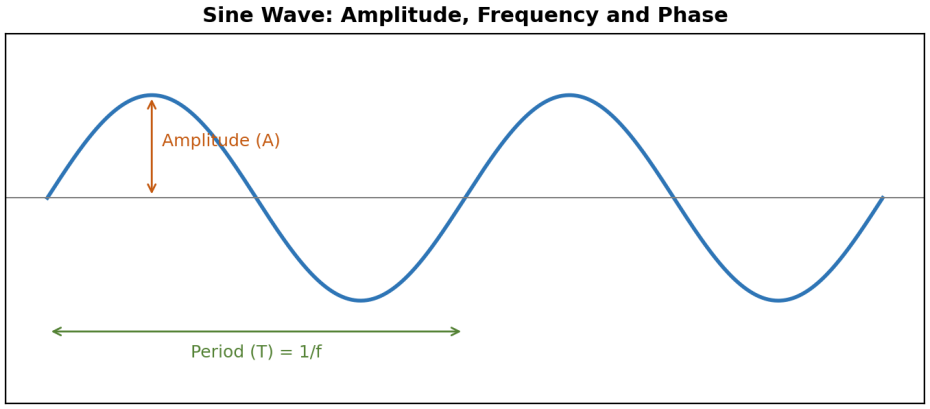

## 6.2 Analog and Digital Signals

Data must be converted into signals to be transmitted. Just as data can be analog or digital, so can the signal used to represent it — and a given type of data is not restricted to only one type of signal (e.g., digital data can be sent as an analog signal via modulation, as discussed in later sections).

| Analog Signal | Digital Signal |
|---|---|
| Has continuous values — can take on infinitely many intensity levels over time. | Has a limited (finite) number of defined levels — most commonly just two (0 and 1). |
| Example: a smooth sine wave. | Example: a square/step waveform switching between fixed voltage levels. |
| **Note:** Simple memory aid — Analog → Continuous. Digital → Discrete. | |

## 6.3 Analog Signals — Periodic vs Non-periodic

An analog signal can be classified as periodic (its pattern repeats at regular intervals — like a heartbeat) or non-periodic / aperiodic (it never repeats exactly). The simplest, most fundamental periodic analog signal is the sine wave, and any composite periodic signal can, by Fourier analysis, be broken down into a sum of simple sine waves of different frequencies.




A sine wave is fully described by three parameters:

| Parameter | Meaning |
|---|---|
| Amplitude (A) | The strength or 'height' of the signal — represents signal energy/power, measured in volts. |
| Frequency (f) | The number of complete cycles (oscillations) completed per second, measured in Hertz (Hz). |
| Phase (φ) | The position of the waveform relative to a fixed reference point in time, usually measured in degrees or radians. |

## 6.4 Period and Frequency

The Period (T) is the time required for a signal to complete exactly one cycle. Frequency and period are reciprocals of each other:

**f = 1 / T and equivalently T = 1 / f**

A high frequency means a short period (rapid oscillation); a low frequency means a long period (slow oscillation). Frequency also describes how fast the signal's value changes with respect to time — since a signal cannot change instantaneously from one value to another, a change in signal value implies a change in frequency.

## 6.5 Wavelength

Wavelength (λ) is the distance a signal travels during one complete period — i.e., the physical spatial length of one cycle of the wave. It depends on both the frequency of the signal and the propagation speed of the medium it travels through:

**λ = propagation speed / frequency = v × T**

Wavelength is particularly important in unguided media (radio, microwave, infrared) and in fiber optics, where the propagation speed of light in the medium (rather than in free space) determines the physical wavelength.

### 6.6 Composite Signals and Fourier Analysis

In real communication systems, a single pure sine wave carries very little useful information (it never changes). Real-world information-carrying signals are composite signals — made up of multiple sine waves of different frequencies, amplitudes, and phases superimposed together.

**Fourier Analysis:** Jean-Baptiste Fourier showed mathematically that any composite periodic signal can be represented as a sum (series) of simple sine waves, each with its own frequency, amplitude, and phase. This mathematical decomposition is what allows engineers to analyze how a complex signal will behave when passed through a channel with limited bandwidth.

### 6.7 Bandwidth

Bandwidth is the range of frequencies contained within a composite signal, or the range of frequencies that a channel can carry. It is calculated as the difference between the highest and lowest frequency components present.

| **Bandwidth = Highest Frequency (fmax) − Lowest Frequency (fmin)** |
|---|

**Worked Example:** If a signal's frequency components range from 1000 Hz to 5000 Hz, then Bandwidth = 5000 − 1000 = 4000 Hz.

Bandwidth is one of the most important concepts in networking — it directly limits how much information (how many bits per second) a channel can theoretically carry, as formalized later by the Nyquist and Shannon theorems.

### 6.8 Digital Signals

Digital signals represent information using a limited, well-defined number of discrete signal levels rather than a continuous range. In the simplest binary case:

- Bit 1 → represented by a positive voltage level.
- Bit 0 → represented by a zero (or negative) voltage level.

A digital signal need not be restricted to only two levels — it can use L discrete levels, where L is a power of 2. The number of bits that can be represented by one such signal level is:

| **Bits per level = $\log_2 L$** |
|---|

| Signal Levels (L) | Bits per Level ($\log_2 L$) |
|---|---|
| 2 | 1 bit |
| 4 | 2 bits |
| 8 | 3 bits |

| | |
|---|---|
| 16 | 4 bits |

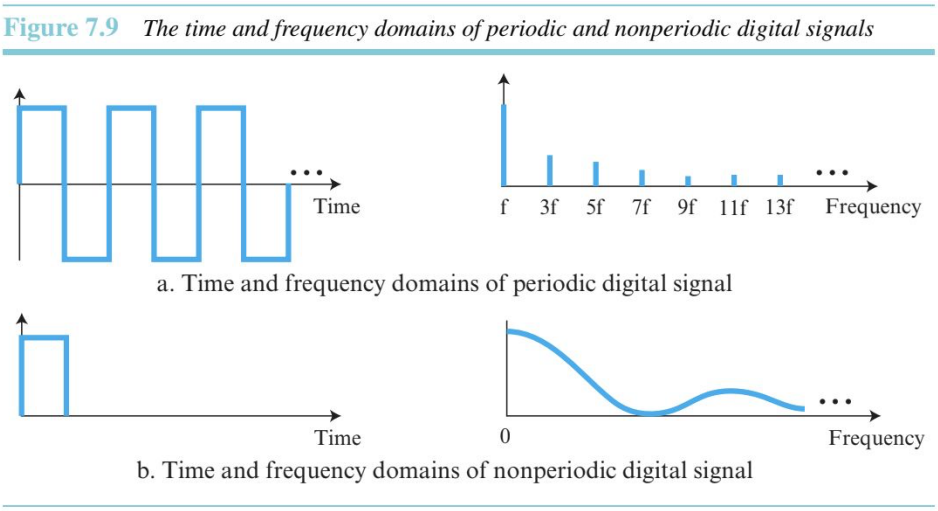



*Figure 7.9 — Time and frequency domains of periodic and nonperiodic digital signals*

## 6.9 Bit Rate and Bit Length

**Bit Rate:** For digital signals, the analog concept of 'frequency' is replaced by bit rate — the number of bits transmitted per second, measured in bits per second (bps).

**Bit Rate = 1000 bits sent in 1 second ⇒ Bit rate = 1000 bps**

**Bit Length:** Bit length is the physical distance a single bit occupies on the transmission medium at a given instant — it is the product of the propagation speed of the medium and the time taken to send one bit (the bit duration). This concept becomes especially important in CSMA/CD-style collision detection on shared media.

## 6.10 A Digital Signal as a Composite Analog Signal

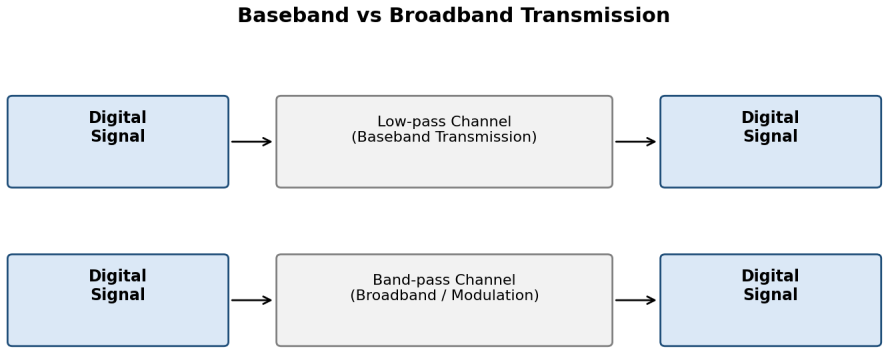
A somewhat counter-intuitive but important result of Fourier analysis is that a 'square' digital signal is, mathematically, itself a composite analog signal made up of a large — theoretically infinite — number of sine-wave frequency components. This is because a digital signal contains sharp vertical transitions (representing very high-frequency components) as well as flat horizontal sections (representing near-zero-frequency components).

- If the digital signal is periodic, its frequency components are discrete (a specific set of frequencies) — but the total bandwidth is still theoretically infinite.
- If the digital signal is non-periodic (the usual case in real data communication, since data bits vary unpredictably), its frequency components are continuous, and its bandwidth is also theoretically infinite.

In practice, however, transmission media and receiving equipment have limited bandwidth, so only a finite range of these frequency components can actually be transmitted and received — this is precisely why signals become distorted, and why the digital-to-analog and digital-to-digital conversion techniques discussed later are so important.

## 6.11 Transmission of Digital Signals: Baseband vs Broadband

A digital signal can be sent over a medium in one of two fundamentally different ways:




**Baseband Transmission:** Sends the digital signal directly through the medium, without converting it into an analog signal first. It requires a low-pass channel — one whose usable frequency range starts at (or near) zero Hz. Baseband transmission is typically used when the entire bandwidth of a dedicated cable is used for one single communication channel, e.g. two computers connected directly by a cable.

**Broadband Transmission (Modulation):** Converts the digital signal into an analog signal before transmission, by using the digital data to modify some property (amplitude, frequency, or phase) of a high-frequency carrier signal. This requires a band-pass channel — one whose usable frequency range does not start at zero. At the receiver, the analog signal is converted back into the original digital data.

**Figure 7.11** *Bandwidth of a band-pass channel*

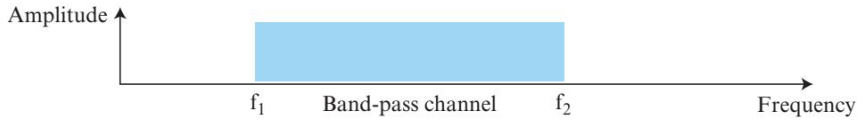



*Figure 7.11 — Bandwidth of a band-pass channel*

**Figure 7.12** *Modulation of a digital signal for transmission on a band-pass channel*

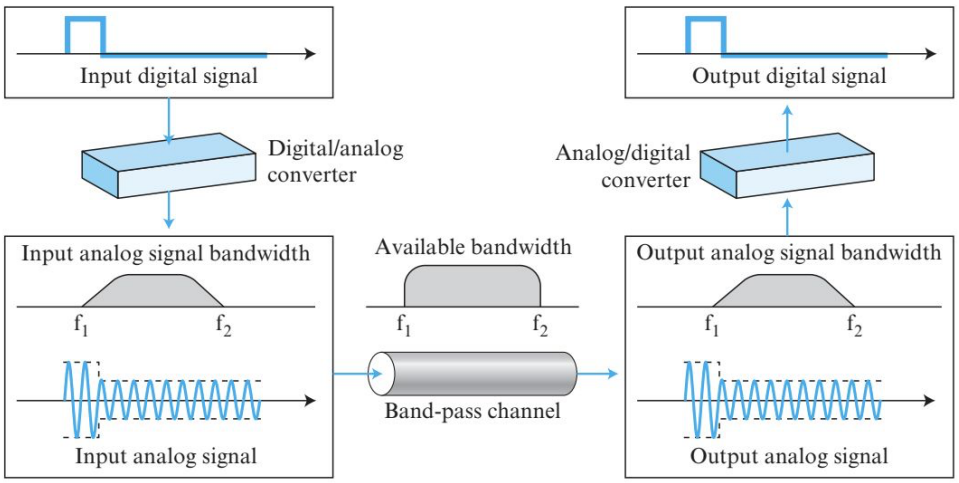



*Figure 7.12 — Modulation of a digital signal for transmission on a band-pass channel*

## 7. Transmission Impairment

No transmission medium is perfect. As a signal travels from sender to receiver, it is inevitably altered, so that the received signal may differ from the transmitted one. This degradation is called transmission impairment. There are three principal causes: attenuation, distortion, and noise.

### 7.1 Attenuation

Attenuation is the loss of signal energy (strength) as it travels through a medium — some signal energy is converted into heat due to the resistance of the medium. To compensate, amplifiers are periodically inserted along long transmission paths to boost the signal strength back up.

Signal strength (gain or loss) is measured in decibels (dB), a logarithmic unit that conveniently compresses very large ratios of power into manageable numbers:

$$\mathbf{dB = 10 \log_{10} (P2 / P1)}$$

where P1 is the power at the beginning of the medium and P2 is the power at the end.

- Negative dB → the signal has been attenuated (weakened).
- Positive dB → the signal has been amplified (strengthened).

**Worked Example:** If the power drops to half its original value (P2 = 0.5 × P1), then dB = 10 $\log_{10}(0.5) \approx -3$ dB. This '−3 dB rule of thumb' (loss of half power ≈ −3 dB) is frequently used in quick calculations.

### 7.2 Distortion

Distortion is a change in the shape or form of a signal, distinct from a simple loss of strength. It typically occurs with composite signals: because a composite signal is made up of multiple frequency components, and different frequencies can travel through a given medium at slightly different speeds, the components may arrive at the destination out of their original phase-alignment. When they are recombined at the receiver, the resulting waveform differs in shape from the one originally sent.

### 7.3 Noise

Noise refers to unwanted electrical or electromagnetic signals that mix with, and interfere with, the intended signal. Several types of noise are commonly identified:

| Type of Noise | Cause |
|---|---|
| Thermal noise | Caused by the random (Brownian) motion of electrons within a conductor; present in all electronic devices. |
| Induced noise | Caused by nearby electrical devices such as motors, appliances, or fluorescent lighting, which act like unintentional transmitting antennas. |
| Crosstalk | Interference caused by signal leakage from a nearby wire or cable carrying a different signal. |
| Impulse noise | Sudden, short-duration, high-energy spikes, caused by sources such as lightning strikes or power-line surges. |

### 7.4 Signal-to-Noise Ratio (SNR)

The Signal-to-Noise Ratio quantifies signal quality by comparing the average power of the desired signal to the average power of the noise present:

**SNR = Average Signal Power / Average Noise Power**

SNR is very frequently expressed in decibels for convenience:

**$SNR_{dB} = 10 \log_{10}(SNR)$**

- High SNR → less relative noise/interference → better signal quality → higher achievable data rate.
- Low SNR → more relative noise/interference → poorer signal quality → lower achievable data rate.

## 8. Data Rate Limits — Nyquist and Shannon

A fundamental question in communications engineering is: given a channel with a certain bandwidth, what is the theoretical maximum rate at which data can be sent reliably? The maximum data rate of a channel depends on three factors: (1) the available bandwidth, (2) the number of signal levels used, and (3) the level of noise present on the channel. Two classical theorems answer this question, depending on whether the channel is noiseless or noisy.

### 8.1 Nyquist Bit Rate — For a Noiseless Channel

For an idealized, noiseless channel, the Nyquist formula gives the theoretical maximum bit rate as a function of bandwidth and the number of signal levels used:

$$\textbf{Bit Rate} = 2 \times B \times \log_2 L$$

where B is the bandwidth of the channel in Hz, and L is the number of discrete signal levels used to represent data.

**Key Insight:** Increasing the number of signal levels (L) can increase the bit rate without increasing the bandwidth — but only up to a point, because in practice, more signal levels are harder for the receiver to distinguish reliably in the presence of any real-world noise.

### 8.2 Shannon Capacity — For a Noisy Channel

Real channels always have some noise. Claude Shannon's channel capacity formula gives the theoretical maximum data rate for a noisy channel, and — importantly — this limit is independent of the number of signal levels used:

$$C = B \times \log_2 (1 + SNR)$$

where C is the channel capacity in bits per second, B is the bandwidth in Hz, and SNR is the (linear, not dB) signal-to-noise ratio.

**Note:** Easy memory rule: Nyquist → Noiseless channel (theoretical, idealized). Shannon → Noisy channel (realistic, accounts for noise). In practice, the achievable rate is always the lower of the two limits.

## 9. Performance Measures of a Network

Beyond the theoretical channel capacity, several practical metrics are used to describe how well a real network actually performs:

### 9.1 Bandwidth

The term bandwidth is used in networking in two related but distinct senses:

- Bandwidth in Hertz (Hz): the range of frequencies that a channel can carry — a property of the analog medium/channel itself.
- Bandwidth in bits per second (bps): the maximum achievable data transmission rate of a link — the more common everyday usage in networking discussions (e.g., 'a 100 Mbps link').

### 9.2 Throughput

Throughput is the actual, measured rate at which data is successfully transmitted over a link in practice — as opposed to bandwidth, which represents the theoretical maximum capacity of the link.

| **Bandwidth = Potential (theoretical) capacity** | **Throughput = Actual achieved data rate** |
|---|---|

**Worked Example:** A link may have a rated bandwidth of 1 Mbps, but due to congestion, protocol overhead, or other bottlenecks, it may only actually carry 200 kbps of real user data — in this case, the throughput is 200 kbps, well below the bandwidth.

### 9.3 Latency (Delay)

Latency is the total time taken for a complete message to travel from the source to the destination. It is the sum of four separate delay components:

| Delay Component | Description |
|---|---|
| Propagation delay | Time for a signal to physically travel across the medium, dependent on distance and propagation speed. |
| Transmission delay | Time required to push all of a packet's bits onto the link, dependent on packet size and bandwidth. |
| Queuing delay | Time a packet spends waiting in a router/switch buffer before being processed, due to congestion. |
| Processing delay | Time a router takes to examine a packet's header and decide how to forward it. |

### 9.4 Jitter

Jitter is the variation in delay experienced by different packets belonging to the same data stream. It is especially problematic for real-time applications such as VoIP calls and video conferencing, where consistent, predictable timing between packets is essential for smooth playback.

**Worked Example:** If Packet 1 experiences a delay of 20 ms, Packet 2 experiences 45 ms, and Packet 3 experiences 40 ms, the variation between these delays is the jitter for that stream. Jitter buffers are used at the receiver to smooth out this variation before playback.

## 10. Digital Transmission (Digital-to-Digital Conversion)

Digital transmission is the process of converting digital or analog source information into a digital signal for transmission. There are two broad conversion paths:

- Digital data → Digital signal — via Digital-to-Digital Conversion (this section).
- Analog data → Digital signal — via Analog-to-Digital Conversion (Section 11).

Digital-to-digital conversion relies on up to three techniques used together, in this typical order:

4. Line coding — always required; converts a bit sequence into a digital signal.
5. Block coding — used when required, to improve synchronization/error-detection.
6. Scrambling — used when required, mainly for long-distance bipolar links.

### 10.1 Line Coding

Line coding is the fundamental, always-required process of converting a sequence of digital bits (0s and 1s) into a digital signal for transmission over a medium. At the sender, an encoder converts bits into a signal; at the receiver, a decoder converts the received signal back into the original bits.

Three important terms are used to characterize line-coding schemes:

| Term | Meaning |
|---|---|
| Bit rate (N) | The number of bits transmitted per second. |
| Baud rate (S) | The number of signal elements (changes) transmitted per second. |
| Bits per signal element (r) | The number of data bits represented/carried by one signal element. |

Line coding schemes are broadly classified into three families: Polar, Bipolar, and Multilevel schemes.

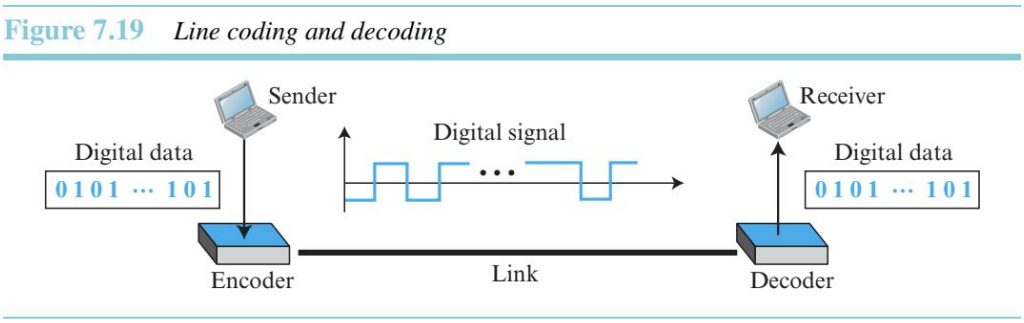



*Figure 7.19 — Line coding and decoding*

### 10.2 Polar Schemes

In polar encoding, the signal uses both positive and negative voltage levels to represent bits.

**NRZ-L (Non-Return-to-Zero, Level):** The voltage level itself directly represents the bit value (e.g. positive voltage = 0, negative voltage = 1, by one common convention).

**NRZ-I (Non-Return-to-Zero, Invert on ones):** Instead of the absolute level, it is the presence or absence of a transition (change) at the start of the bit interval that carries meaning: a transition represents bit 1, and no transition represents bit 0.

**Note:** A key weakness of both NRZ schemes: a long run of consecutive 0s (in NRZ-I) or a long run of identical bits (in NRZ-L) produces no signal transitions for an extended time, making it very difficult for the receiver's clock to stay synchronized with the sender's clock. This 'synchronization problem' motivates the later encoding schemes (RZ, Manchester, Block Coding).

**RZ (Return-to-Zero):** Uses three signal levels — positive, negative, and zero. The signal always returns to the zero level partway through every bit interval, guaranteeing at least one transition per bit and thus solving the synchronization problem. Advantage: reliable synchronization. Disadvantage: requires roughly double the bandwidth of NRZ, since more transitions occur per second.

**Manchester:** Combines the ideas of NRZ-L and RZ. There is always a transition exactly in the middle of every bit interval, which is used both to represent the bit value and to provide clock synchronization simultaneously. (Convention: low-to-high transition = 1, high-to-low = 0, or vice versa depending on the standard used — this is the encoding historically used by classic 10 Mbps Ethernet.)

**Differential Manchester:** Also has a transition in the middle of every bit (for synchronization), but the actual bit value is determined by whether or not there is an additional transition at the very beginning of the bit interval: a transition at the start represents bit 0, while no transition at the start represents bit 1.

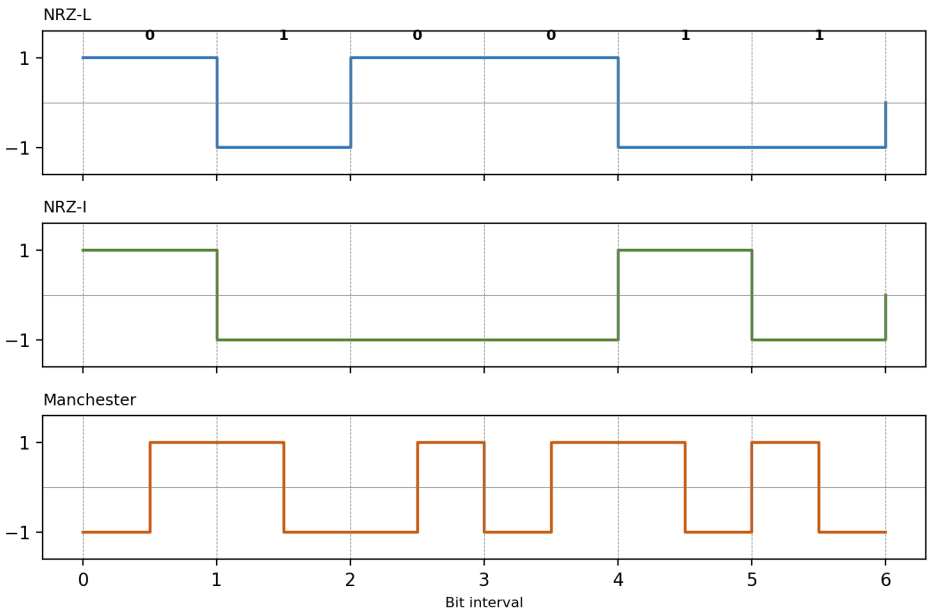


Figure 7.20 *Polar schemes*

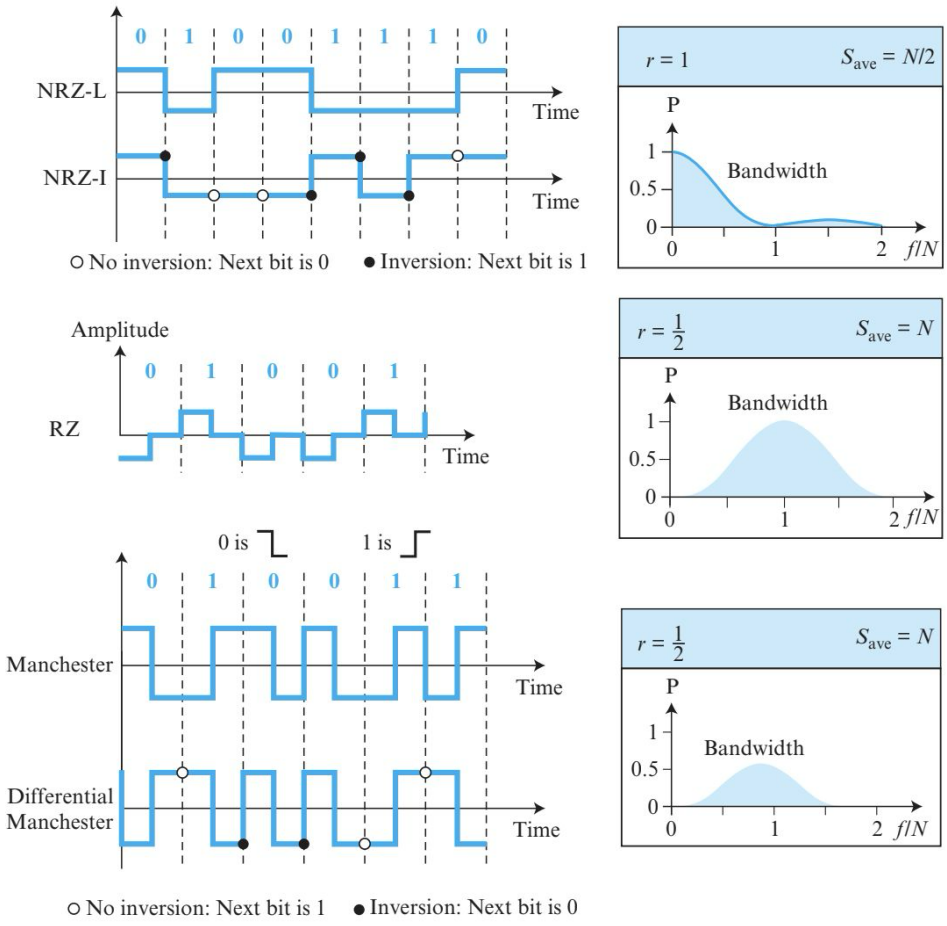



*Figure 7.20 — Polar schemes (NRZ-L, NRZ-I, RZ, Manchester, Differential Manchester)*

## 10.3 Bipolar Schemes

Bipolar encoding uses three voltage levels — positive, negative, and zero — but unlike RZ, the zero level is reserved specifically for one bit value, while the other bit value alternates between positive and negative.

**AMI (Alternate Mark Inversion):** Bit 0 is always represented by zero voltage. Bit 1 is represented by alternating positive and negative voltage pulses each time a 1 occurs (so consecutive 1s never use the same polarity twice in a row). For example, three consecutive 1-bits would be encoded as +V, −V, +V.

**Pseudoternary:** The exact opposite convention of AMI — bit 1 is represented by zero voltage, while bit 0 alternates between positive and negative voltage.

Bipolar/AMI encoding has two significant practical advantages over polar schemes: a narrower required bandwidth, and — because the positive and negative pulses for 1s balance each other out over time — it produces no net DC

(direct current) component, which is important because DC components cannot pass through many types of transformers and AC-coupled equipment commonly used in long-distance transmission lines.

**Figure 7.21** *Bipolar schemes: AMI and pseudoternary*

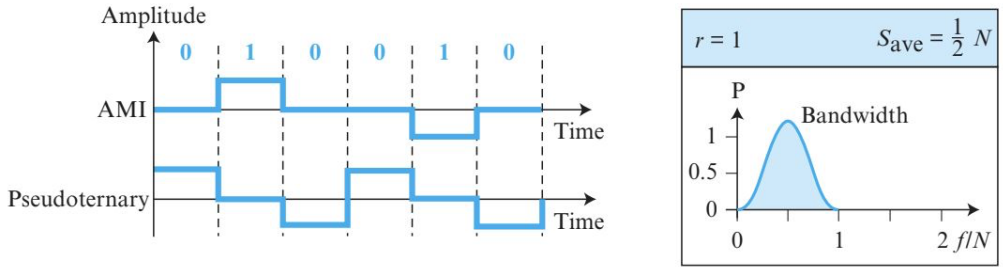



*Figure 7.21 — Bipolar schemes: AMI and pseudoternary*

## 10.4 Multilevel Schemes

Multilevel encoding schemes are designed to increase the effective data rate (or, equivalently, reduce the bandwidth needed for a given data rate) by encoding multiple bits into a single signal element rather than one. The general notation used is:

| **mBnL** | |
|---|---|
| **Symbol** | **Meaning** |
| m | Number of input (data) bits grouped together. |
| B | Indicates binary (source) data. |
| n | Number of signal elements produced per group. |
| L | Number of distinct signal levels used. |

**2B1Q (2 Binary, 1 Quaternary):** Every group of 2 input bits is encoded as a single signal element chosen from 4 possible voltage levels (quaternary). This effectively doubles the data rate for a given baud rate compared to a 2-level scheme. Used historically in ISDN and some DSL variants.

**8B6T (8 Binary, 6 Ternary):** Every group of 8 input bits is encoded as 6 signal elements chosen from 3 voltage levels (ternary). The extra redundant patterns this produces are useful for synchronization and simple error detection, and this scheme is used in 100BASE-4T Ethernet.

Figure 7.22 *Multilevel: 2B1Q and 8B6T*

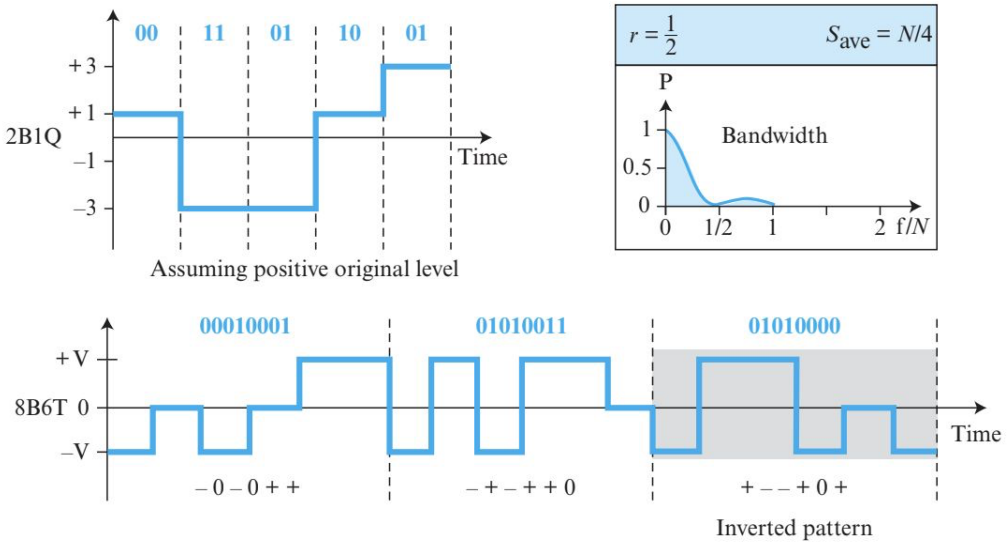



*Figure 7.22 — Multilevel schemes: 2B1Q and 8B6T*

## 10.5 Block Coding

Block coding is used in addition to line coding (not instead of it) to add redundant bits to the original data before it is line-coded. These extra bits serve important purposes:

- Maintain synchronization between sender and receiver by guaranteeing frequent enough signal transitions.
- Provide basic error detection capability at the receiver.
- Improve the overall performance and reliability of the chosen line-coding scheme.

Block coding is denoted as mB/nB, meaning a block of m input bits is systematically substituted with a corresponding block of n output bits, where n > m (the extra bits are the added redundancy). Block coding is performed in three steps:

7. Division — the original bit stream is divided into groups of m bits each.
8. Substitution — each m-bit group is looked up and replaced by its corresponding, pre-defined n-bit code.
9. Combination — all the resulting n-bit groups are concatenated together to form the new, longer bit stream that is then passed to the line-coding stage.

Figure 7.23 *Block coding concept*

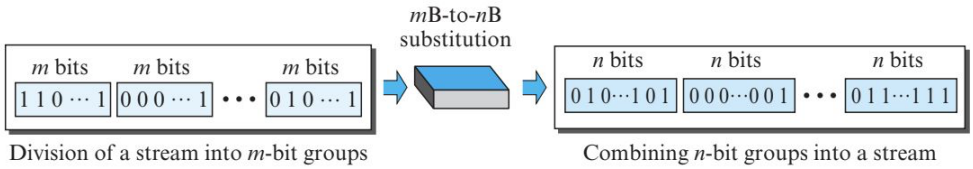



*Figure 7.23 — Block coding concept (mB/nB substitution)*

## 10.6 4B/5B Coding

4B/5B was specifically designed to solve the synchronization weakness of NRZ-I (long runs of 0s producing no transitions). It converts every group of 4 input bits into a carefully chosen 5-bit output code, selected such that no output code contains a long run of zeros.

Specifically, each 5-bit code is designed to have:

- At most one leading zero, and
- At most two trailing zeros.

Because of this careful design, even when different 5-bit codes are concatenated back-to-back, the combined stream can never contain more than three consecutive zeros — which is short enough for NRZ-I's receiver clock to stay synchronized. 4B/5B is typically combined with NRZ-I line coding in practice (e.g., in 100BASE-TX Fast Ethernet and FDDI).

**NRZ-I signal → decode into 5-bit groups → strip the extra bit → recover original 4-bit data**

Figure 7.24 *Using block coding 4B/5B with NRZ-I line coding scheme*

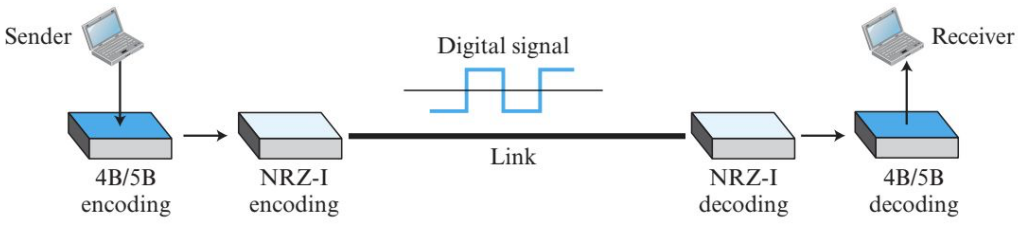



*Figure 7.24 — Using block coding 4B/5B with the NRZ-I line coding scheme*

## 10.7 8B/10B Coding

8B/10B extends the same idea, converting every 8 data bits into a 10-bit output code, providing even better error detection and synchronization performance than 4B/5B. It is used in high-speed standards such as Gigabit Ethernet, USB 3.0, and PCI Express (older generations).

The 8-bit input is internally split into two smaller groups that are each block-coded separately, then combined:

**5B/6B + 3B/4B = 8B/10B (5 bits → 6-bit code, combined with 3 bits → 4-bit code)**

**Disparity Controller:** 8B/10B additionally uses a running disparity controller, which tracks whether the transmitted codes so far have contained more 0s than 1s, or more 1s than 0s. If necessary, it complements (inverts) the bits of the current output code (0→1, 1→0) to correct any imbalance and keep the overall stream DC-balanced.

Since 10 bits can represent $2^{10} = 1024$ possible patterns, but only $2^8 = 256$ are needed to represent the original data, there are:

**$2^{10} - 2^8 = 768$ redundant/unused code patterns**

These unused patterns are deliberately reserved and used for error detection and disparity checking.

Figure 7.25 *8B/10B block encoding*

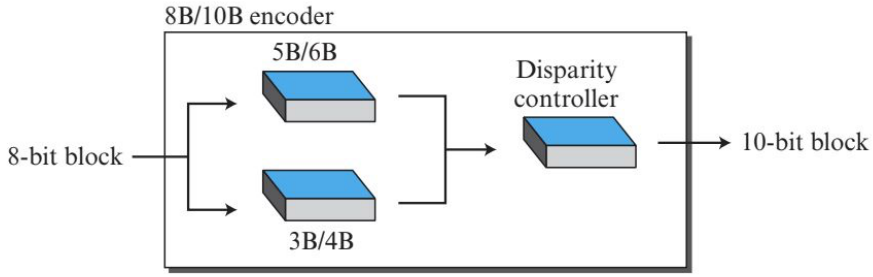



*Figure 7.25 — 8B/10B block encoding*

## 10.8 Scrambling

Scrambling is an alternative (or complementary) technique to block coding, used mainly with bipolar AMI encoding for long-distance transmission lines (such as T-1 telephone lines).

AMI has two very attractive properties: a narrow required bandwidth, and no DC component. However, it shares NRZ's core weakness — a long run of 0s produces no voltage transitions at all (since 0 is always encoded as zero voltage in AMI), causing loss of synchronization.

A good solution needs to satisfy all of the following simultaneously:

- Avoid long sequences of consecutive zeros.
- Maintain reliable synchronization.
- NOT increase the number of bits transmitted (unlike block coding, which adds redundant bits).
- Preserve AMI's existing advantages (narrow bandwidth, no DC component).

This technique — substituting specific 'violation' patterns in place of long zero-runs, without adding extra bits — is called scrambling.

Figure 7.26 *AMI used with scrambling*

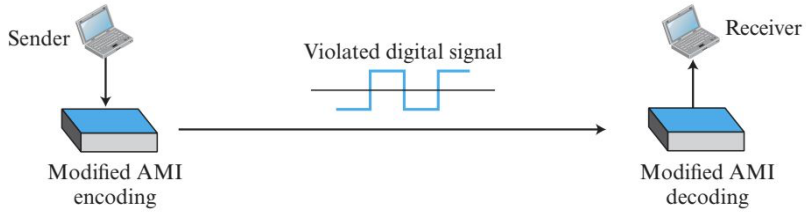



*Figure 7.26 — AMI used with scrambling*

## 10.9 Block Coding vs Scrambling

| Block Coding | Scrambling |
|---|---|
| Adds extra (redundant) bits to the data. | Does not add any extra bits. |
| Increases the total number of bits transmitted. | The bit rate remains exactly the same. |
| Performed before line coding, as a separate stage. | Performed as part of the encoding process itself. |
| Examples: 4B/5B, 8B/10B | Examples: B8ZS, HDB3 |

## 10.10 B8ZS Coding (Bipolar with 8-Zero Substitution)

B8ZS is a scrambling technique commonly used in North America. Whenever exactly eight consecutive zero-bits occur in the data, they are substituted with the following special 8-bit pattern:

**000VB0VB**

- B = a Bipolar pulse that still follows the normal AMI alternation rule.
- V = a Violation pulse that deliberately breaks (violates) the normal AMI alternation rule — this deliberate rule violation is exactly what allows the receiver to recognize that a substitution has occurred and correctly recover the original eight zeros.

Important properties of B8ZS:

- The eight zero-bits are replaced by eight new pulses/bits, so the bit rate is unchanged.
- The deliberately inserted transitions restore synchronization.
- Positive and negative pulses remain balanced overall, so the desirable 'no DC component' property of AMI is preserved.

**Figure 7.27** *Two cases of B8ZS scrambling technique*

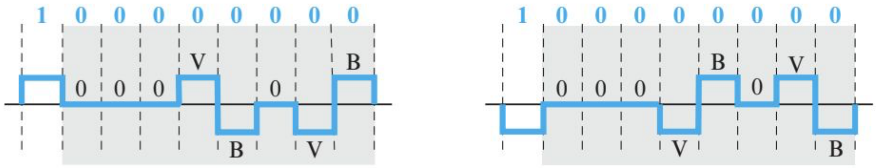



*Figure 7.27 — Two cases of the B8ZS scrambling technique*

## 10.11 HDB3 Coding (High-Density Bipolar 3-Zero)

HDB3 is the scrambling technique commonly used outside North America (particularly in Europe, following ITU-T standards). Whenever four consecutive zero-bits occur, they are substituted with one of two special patterns:

| **000V or B00V** |
|---|

The choice between these two substitution patterns depends on the number of non-zero pulses transmitted since the previous substitution:

- If that count of non-zero pulses is odd → use 000V.
- If that count of non-zero pulses is even → use B00V.

This rule ensures the overall stream maintains the required balance between positive and negative pulses (keeping the average DC level at zero), while still guaranteeing frequent enough transitions for synchronization.

Figure 7.28 *Different situations in HDB3 scrambling technique*

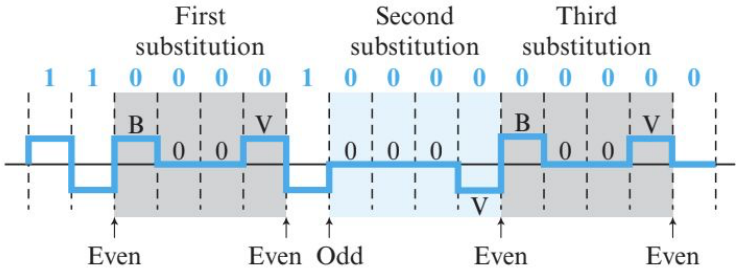


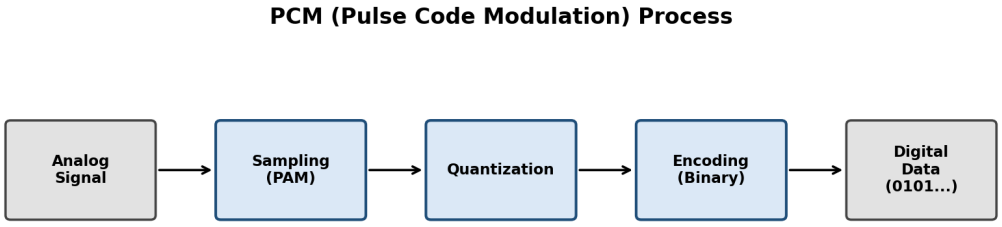

*Figure 7.28 — Different situations in the HDB3 scrambling technique*

## 11. Analog-to-Digital Conversion

Many real-world signals — such as the voice from a microphone, or an image from a camera — originate as analog signals. Since digital signals are generally less affected by noise and are easier to regenerate/reproduce exactly, analog signals are frequently converted into digital data before storage or transmission. This conversion process is called digitization. The two classic techniques are Pulse Code Modulation (PCM) and Delta Modulation (DM). After digitization, the resulting digital data can be further converted into a digital signal using the digital-to-digital line-coding techniques already covered in Section 10.

### 11.1 Pulse Code Modulation (PCM)

PCM is the most common and widely used technique for converting an analog signal into digital data. PCM consists of three sequential steps:




10. Sampling — the analog signal's amplitude is measured at regular, fixed time intervals.
11. Quantization — each sampled amplitude value is approximated (rounded) to the nearest one of a fixed, finite number of discrete levels.
12. Encoding — each quantized value is converted into a corresponding binary codeword.

### 11.2 Sampling

Sampling means measuring (and momentarily 'freezing') the amplitude of an analog signal at regular intervals in time. If samples are taken every Ts seconds, then Ts is called the sampling interval, and the number of samples taken every second is called the sampling rate or sampling frequency, denoted fs:

**fs = 1 / Ts**

The sampling process itself can be thought of as a form of analog modulation called Pulse Amplitude Modulation (PAM), since the amplitude of each output pulse directly represents the corresponding sampled value. The most commonly used practical technique is sample-and-hold, in which the sampled amplitude value is held constant (as a flat-top pulse) for a short duration — this makes the subsequent quantization step easier to implement in hardware.

**Figure 7.30** *Three different sampling methods for PCM*

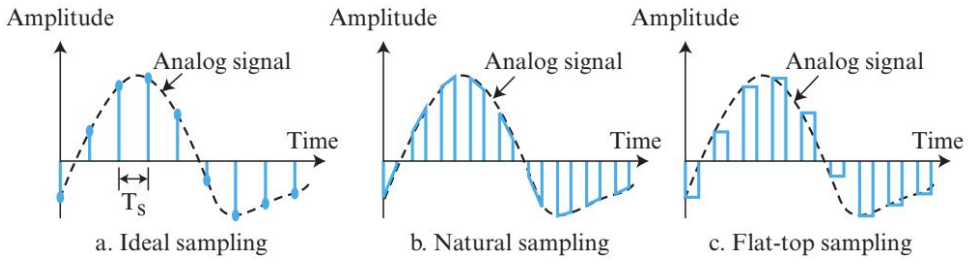



*Figure 7.30 — Three different sampling methods for PCM*

## 11.3 The Nyquist Sampling Theorem

A critical question is: how frequently must an analog signal be sampled so that it can later be perfectly reconstructed from those samples? The Nyquist Sampling Theorem provides the answer: to be able to accurately reproduce the original analog signal, the sampling rate must be at least twice the highest frequency component present in that signal.

**fs ≥ 2 × fmax**

This minimum required sampling rate (2 × fmax) is called the Nyquist rate. Sampling below this rate causes a distortion effect called aliasing, where high-frequency components are misrepresented as false lower-frequency components in the reconstructed signal — this is why a low-pass anti-aliasing filter is normally applied to the analog signal before sampling.

**Worked Example — Human Voice:** Human voice signals typically contain frequency components up to about 4000 Hz. Applying the Nyquist theorem:

**fs = 2 × 4000 = 8000 samples per second**

This is exactly why standard telephone-quality voice digitization uses a sampling rate of 8000 samples/second — a widely used, frequently examined figure.

**Figure 7.31** *Nyquist sampling rate for low-pass and bandpass signals*

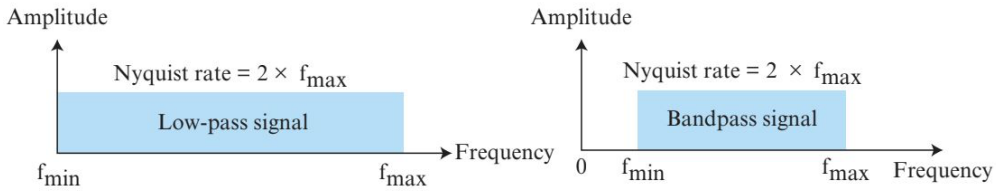



*Figure 7.31 — Nyquist sampling rate for low-pass and bandpass signals*

## 11.4 Quantization

After sampling, each sample's amplitude may in principle take on any value within a continuous range — but this continuous range must be converted into one of a fixed, finite number of levels before it can be represented digitally. This approximation process is called quantization.

The amplitude range from Vmin to Vmax is divided into L quantization levels. Each level represents a small sub-range of amplitudes, and each actual sample value is approximated (rounded) to whichever quantization level is nearest to it. The size (step) of each quantization interval is:

$$\Delta = (V_{max} - V_{min}) / L$$

**Worked Example:** If a signal ranges from −20 V to +20 V and 8 quantization levels are used:

$$\Delta = (20 - (-20)) / 8 = 40 / 8 = 5 \text{ V}$$

This means the full amplitude range is divided into eight 5-volt-wide bands, and every sample is assigned to whichever of the eight levels its actual value falls into.

## 11.5 Quantization Error and Quantization Noise

Because quantization is an approximation process, the original (continuous) sample value and its quantized (rounded) value will generally not be exactly identical. This difference is called the quantization error. Since each sample is rounded to the nearest level, the maximum possible quantization error is bounded by half of the quantization step:

$$-\Delta/2 \leq \text{Error} \leq \Delta/2$$

Increasing the number of quantization levels (L) makes each interval Δ smaller, reducing the maximum quantization error and improving the fidelity of the reconstructed signal — but this comes at the cost of requiring more bits per sample to encode. The resulting signal-to-noise ratio contribution from quantization alone is approximately:

$$SNR_{dB} \approx 6.02 \times nb + 1.76 \text{ dB}$$

where nb is the number of bits used to represent each sample. This formula shows that each additional bit per sample improves the SNR by roughly 6 dB — a useful rule of thumb in audio engineering (this is why, for example, 16-bit CD audio has a noticeably higher dynamic range/SNR than older 8-bit digital audio).

### 11.6 Quantization Levels and Bits per Sample

The number of bits required to represent each quantized sample depends directly on the number of quantization levels, L:

$$nb = \log_2 L$$

**Worked Example:** If L = 8 levels, then nb = $\log_2 8$ = 3 bits per sample. A quantization level of '2' might therefore be encoded in binary as 010, while level '5' would be encoded as 101.

In practice, standard telephone-quality audio digitization typically uses 256 quantization levels (8 bits/sample), while video digitization may require thousands of levels because visual data contains far greater amplitude variation and requires much higher fidelity.

### 11.7 Encoding and the PCM Bit Rate

The final PCM step, encoding, simply represents each quantized sample's level number using its corresponding binary codeword (as derived above). The overall data rate produced by the whole PCM process — the PCM bit rate — is the product of the sampling rate and the number of bits used per sample:

$$\text{Bit Rate} = \text{Sampling Rate (fs)} \times \text{Bits per Sample (nb)}$$

**Worked Example — Human Voice (continued):** Using fs = 8000 samples/second and nb = 8 bits/sample (256 levels):

$$\text{Bit Rate} = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

This is precisely why standard, uncompressed digital telephone voice channels (as used in ISDN and the classic digital telephone network) operate at exactly 64 kbps — a figure worth memorizing, as it is a frequently examined numerical result.

### 11.8 Recovery of the Original Signal (PCM Decoding)

At the receiving end, a PCM decoder converts the received binary codewords back into their corresponding quantized amplitude values. Because these recovered values jump directly between discrete levels, they initially form a rough, 'staircase-like' signal. A low-pass filter is then applied to smooth out this staircase shape and reconstruct an approximation of the original continuous analog signal. Provided the original signal was sampled at or above the Nyquist rate, and a sufficiently large number of quantization levels was used, the reconstructed signal will closely approximate the original with acceptable fidelity.

Figure 7.33 *Components of a PCM decoder*

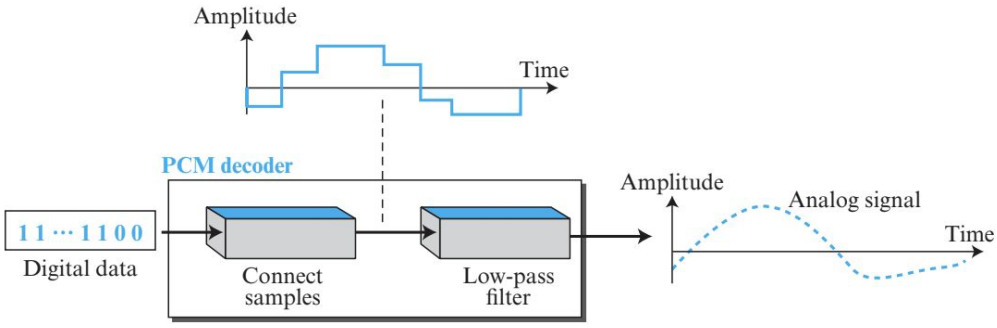



*Figure 7.33 — Components of a PCM decoder*

## 11.9 PCM Bandwidth Requirement

Digitizing an analog signal significantly increases the bandwidth required for transmission, because the original low-bandwidth analog signal is converted into a much higher-rate digital bit stream. The minimum bandwidth of the resulting digital signal is approximately:

**Bmin = nb × Banalog**

where nb is the number of bits used per sample, and Banalog is the bandwidth of the original analog signal. This clearly shows the fundamental trade-off in PCM design: using more bits per sample improves reconstruction accuracy (higher SNR), but also proportionally increases the required digital bit rate and transmission bandwidth.

## 11.10 Delta Modulation (DM)

Delta Modulation is a simpler alternative to PCM. Where PCM records and transmits the complete, actual amplitude value of every single sample, Delta Modulation records only the change (delta) in amplitude between the current sample and the immediately preceding one. Consequently, DM does not need to send a full multi-bit code word for every sample — it needs to send only a single bit per sample, simply indicating whether the signal amplitude has increased or decreased relative to the previous sample.

**PCM asks: "What is the amplitude?"  DM asks: "Did the amplitude go up or down?"**

Because of this much simpler encoding rule, DM requires significantly less complex transmitter/receiver hardware than full PCM. However, its major drawback is that it can introduce substantial tracking errors ('slope overload') when the input signal changes amplitude very rapidly, since the output can only step up or down by one fixed small increment per sample interval.

| Stage | PCM Flow | DM Flow |
|---|---|---|
| Encoding (transmitter) | Analog signal → Sample → Quantize → Encode → Bits | Analog signal → Sample → Compare to previous sample → Output 1 bit (up/down) |
| Decoding (receiver) | Bits → Decode → Staircase signal → Low-pass filter → Analog signal | Bits → Reconstruct staircase from up/down steps → Low-pass filter → Analog signal |

Figure 7.34 *The process of delta modulation*

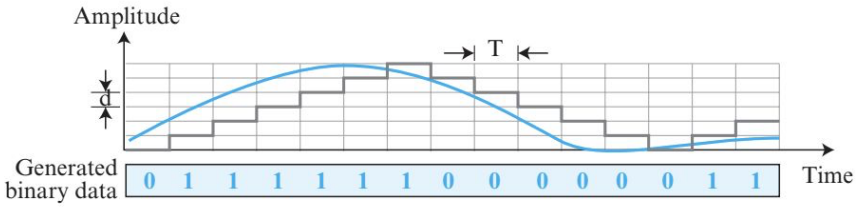



*Figure 7.34 — The process of delta modulation*

## 12. Analog Transmission

Analog transmission is used whenever the only communication channel available is a bandpass channel — one whose usable frequency range does not start at (or near) zero Hz. Since digital (baseband) transmission generally requires a low-pass channel, digital data must first be converted into an analog signal before it can be sent over such a bandpass channel; this conversion is called digital-to-analog conversion. Similarly, when an existing analog signal needs to be shifted into a different frequency range to suit a bandpass channel, it undergoes analog-to-analog conversion.

### 12.1 Digital-to-Analog Conversion — Overview

Digital-to-analog conversion is the process of converting digital data (0s and 1s) into an analog signal suitable for transmission over a bandpass channel. This is achieved by taking a high-frequency sinusoidal carrier signal and modifying (modulating) one or more of its three basic characteristics according to the incoming digital data:

| Characteristic | Description | Modulation technique that varies it |
|---|---|---|
| Amplitude | The height/strength of the wave. | ASK — Amplitude Shift Keying |
| Frequency | The number of cycles completed per second. | FSK — Frequency Shift Keying |
| Phase | The position of the waveform relative to time. | PSK — Phase Shift Keying |
| Amplitude + Phase | Both varied together. | QAM — Quadrature Amplitude Modulation |

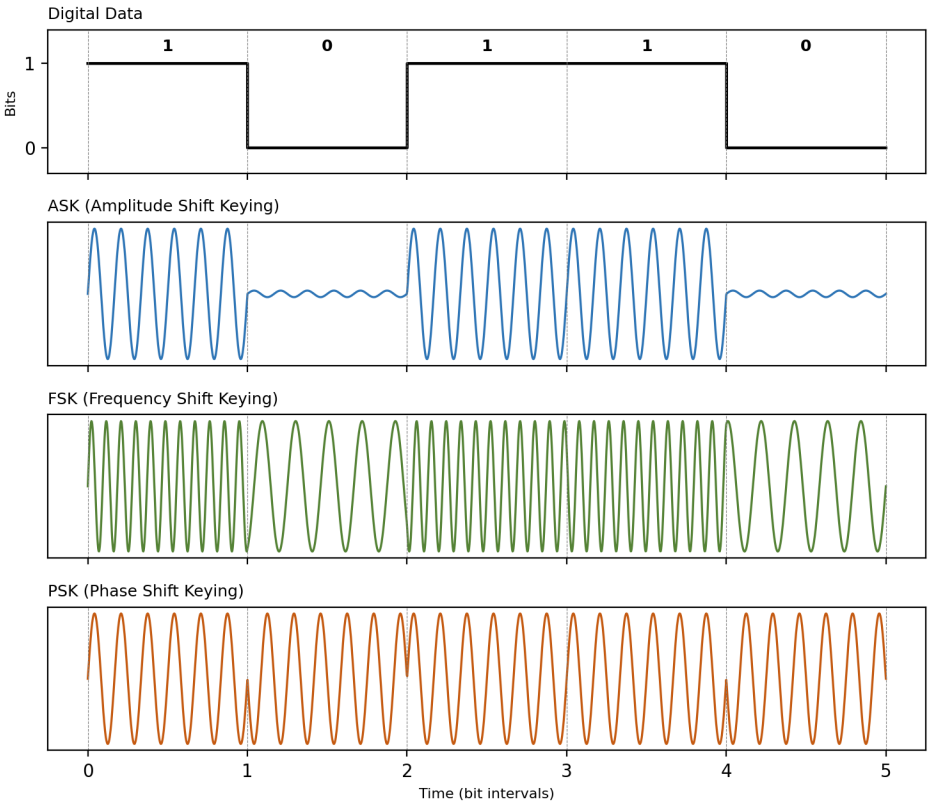



**Figure 7.35** *Digital-to-analog conversion*

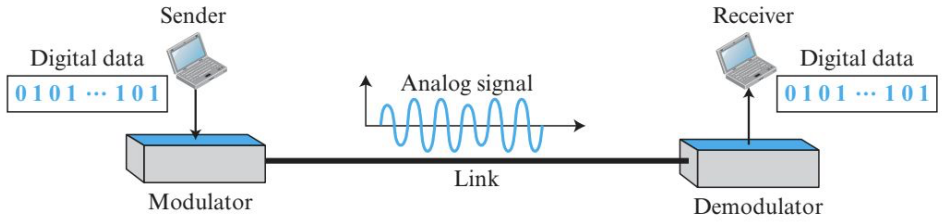



*Figure 7.35 — Digital-to-analog conversion*

## 12.2 Amplitude Shift Keying (ASK)

In ASK, the amplitude of the carrier signal is varied according to the digital data being sent, while the frequency and phase of the carrier remain constant throughout.

- Bit 1 → represented by a higher (full) amplitude.
- Bit 0 → represented by a zero or much lower amplitude.

The simplest possible form of ASK is called Binary ASK (BASK), also commonly known as On-Off Keying (OOK) — in this scheme, one signal level literally has zero amplitude (carrier completely 'off'), while the other level is the

full carrier amplitude ('on'). For a bit sequence such as 1 0 1 1 0, the carrier is effectively switched: ON → OFF → ON → ON → OFF.

**ASK = Change the amplitude to represent bits**

**ASK Bandwidth:** The bandwidth required for ASK depends on the signal (baud) rate S and a filtering/modulation factor d (where $0 \leq d \leq 1$):

**$B = (1 + d) \times S$**

This means the minimum practical bandwidth required is approximately equal to S (the baud rate), and the maximum can approach 2S. An important practical advantage of ASK (and the other keying techniques) is that the carrier frequency fc can be freely chosen to suit whatever bandpass channel is actually available.

**Multilevel ASK:** Rather than using only two amplitude levels, ASK can use multiple discrete amplitude levels — 4 levels carry 2 bits per signal element, 8 levels carry 3 bits, 16 levels carry 4 bits, and so on — allowing more data to be sent per signal element. In practice, however, high-order amplitude-based modulation is almost always implemented as QAM rather than pure ASK, since QAM combines amplitude changes with phase changes for better noise performance.

Figure 7.36 *Binary amplitude shift keying*

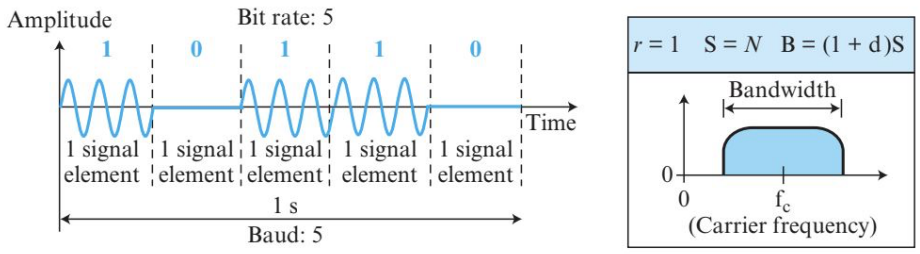



*Figure 7.36 — Binary amplitude shift keying*

## 12.3 Frequency Shift Keying (FSK)

In FSK, the frequency of the carrier signal is varied according to the digital data, while the amplitude and phase remain constant. In Binary FSK (BFSK), two distinct carrier frequencies are used:

- Frequency f1 → represents bit 0.
- Frequency f2 → represents bit 1.

**FSK = Change the frequency to represent bits**

**Note:** Easy way to remember: imagine a person's voice pitch — bit 0 corresponds to a 'slow/low' tone, and bit 1 corresponds to a 'fast/high' tone, while the loudness (amplitude) stays roughly the same throughout.

**FSK Bandwidth:** The bandwidth required for FSK must account for both the signal rate and the separation between the two carrier frequencies:

$$B = (1 + d) \times S + 2\Delta f$$

where $2\Delta f$ represents the separation between the two carrier frequencies used. For the receiver to reliably distinguish the two frequencies, this separation must be sufficiently large.

**Multilevel FSK (MFSK):** Instead of only two carrier frequencies, many distinct frequencies can be used — 4 frequencies carry 2 bits per signal element, 8 frequencies carry 3 bits, 16 frequencies carry 4 bits, and so on. The different frequencies must remain sufficiently well-separated for the receiver to correctly distinguish between them.

## 12.4 Phase Shift Keying (PSK)

In PSK, the phase of the carrier signal is varied according to the digital data, while the amplitude and frequency remain constant.

**PSK = Change the phase to represent bits**

**Binary PSK (BPSK):** The simplest form of PSK, using just two phase values — typically 0° and 180°. For example, bit 0 might be assigned the 0° phase, and bit 1 the 180° phase; the carrier's phase simply flips by 180° whenever the bit value changes.

**Why PSK outperforms ASK in the presence of noise:** In ASK, the receiver identifies each bit primarily by measuring the signal's amplitude — but amplitude is precisely the property most easily distorted by noise and attenuation. In PSK, the receiver instead identifies each bit by measuring the signal's phase, and random amplitude fluctuations caused by noise have comparatively little direct effect on the measured phase. As a result, BPSK is generally considerably more robust/less susceptible to amplitude-related noise than ASK.

**Quadrature PSK (QPSK):** An enhanced form of PSK that transmits 2 bits per signal element instead of just 1, by using four distinct phase states instead of two, since $2^2 = 4$:

| Bit pair | Phase |
|---|---|
| 00 | 0° |
| 01 | 90° |
| 10 | 180° |
| 11 | 270° |

The key advantage of QPSK is that 2 bits are transmitted per signal element, which means the required baud rate (and hence bandwidth) for a given target bit rate can be halved compared to BPSK — a very significant practical efficiency gain, which is why QPSK and its higher-order relatives are widely used in real-world wireless standards.

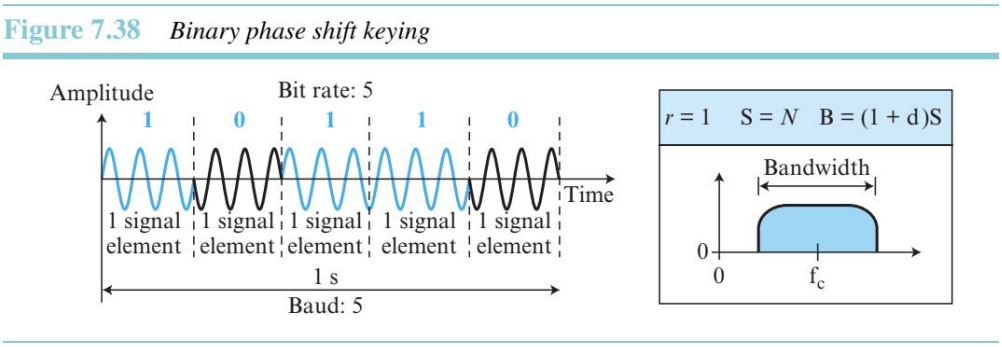



*Figure 7.38 — Binary phase shift keying*

## 12.5 The Constellation Diagram

A constellation diagram is a standard graphical technique used to represent the set of possible signal elements used in modulation schemes such as PSK, ASK, and QAM, making it easy to visually compare different schemes and analyze their noise resilience.

- X-axis: represents the In-phase (I) component of the signal.
- Y-axis: represents the Quadrature (Q) component of the signal (i.e., a component shifted 90° out of phase with the I component).

Every possible transmitted signal state is plotted as a single point on this diagram. From the position of a point, key signal properties can be read off directly:

| Diagram property | Corresponding signal property |
|---|---|
| Distance of the point from the origin (centre) | Amplitude |
| Angle of the point relative to the X-axis | Phase |
| Position along the X-axis | I (in-phase) component |
| Position along the Y-axis | Q (quadrature) component |

**Note:** Memory rule: Distance from centre = Amplitude. Angle from the X-axis = Phase.

Figure 7.39 *Concept of a constellation diagram*

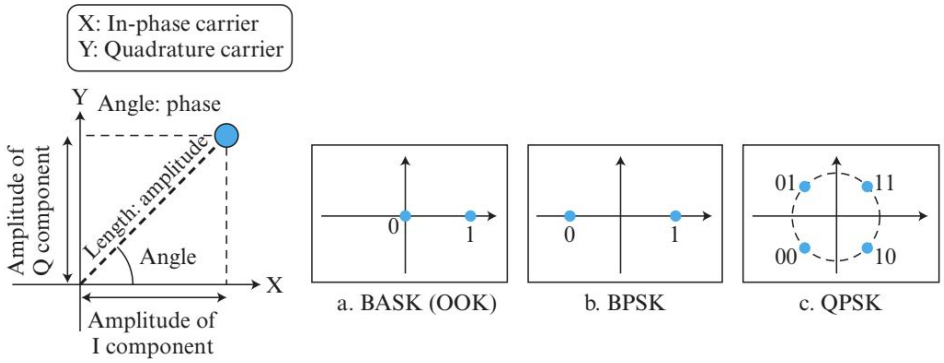



*Figure 7.39 — Concept of a constellation diagram*

## 12.6 Quadrature Amplitude Modulation (QAM)

QAM combines the advantages of both ASK and PSK — it varies both the amplitude and the phase of the carrier simultaneously, allowing significantly more bits to be packed into each transmitted signal element than either technique could achieve alone.

**QAM = Change amplitude + Change phase**

QAM is implemented using two separate carrier signals of the same frequency but 90° out of phase with each other:

- The In-phase carrier (I).
- The Quadrature carrier (Q), shifted 90° relative to I.

By varying the amplitudes with which each of these two carriers is combined, QAM can generate a large number of distinct, distinguishable signal points ('constellation points'). Common QAM variants include:

| Scheme | Bits per signal element |
|---|---|
| 4-QAM | 2 bits/signal element |
| 16-QAM | 4 bits/signal element |
| 64-QAM | 6 bits/signal element |
| 256-QAM | 8 bits/signal element |

As the number of constellation points (signal combinations) increases, more bits can be packed into each signal element, increasing spectral efficiency (bits per second per Hz of bandwidth) — but the constellation points also become more closely spaced, making the scheme more susceptible to noise-induced errors. Because of this excellent trade-off between efficiency and (with a large enough SNR) reliability, QAM is very widely used in modern high-speed systems including cable modems, DSL, and Wi-Fi.

## 12.7 Summary: ASK vs FSK vs PSK vs QAM

| Technique | What changes? | What remains constant? |
|---|---|---|
| ASK | Amplitude | Frequency and phase |
| FSK | Frequency | Amplitude and phase |
| PSK | Phase | Amplitude and frequency |
| QAM | Amplitude + Phase | Frequency |

## 12.8 Analog-to-Analog Conversion (Analog Modulation)

Analog-to-analog conversion, also called simply analog modulation, is the process of representing one analog information signal using a different analog signal (typically at a different, higher frequency).

A natural question arises: if the original information signal is already analog, why would we need to modulate it further? The answer is that most real communication channels are bandpass channels that occupy a specific, limited frequency band. For example, in radio broadcasting, many independent radio stations each generate a low-frequency, low-pass audio signal. If every station transmitted its raw audio signal directly (all occupying the same low-frequency range), all their transmissions would hopelessly interfere with one another. The solution is that each station is instead assigned its own unique carrier frequency, and its audio signal is shifted up onto that carrier frequency via modulation — this is what allows many different stations to share the airwaves (the electromagnetic spectrum) simultaneously without interfering.

There are three basic analog-to-analog modulation techniques, directly paralleling the digital-to-analog techniques already discussed:

13. Amplitude Modulation (AM)
14. Frequency Modulation (FM)
15. Phase Modulation (PM)

## 12.9 Amplitude Modulation (AM)

In AM, the amplitude of the high-frequency carrier signal is varied continuously in proportion to the instantaneous amplitude of the original, low-frequency analog information signal, while the carrier's frequency and phase remain constant. The shape traced out by the varying peaks of the carrier — called the envelope — mirrors the shape of the original information signal.

- If the information signal's amplitude becomes stronger → the carrier's amplitude becomes correspondingly larger.
- If the information signal's amplitude becomes weaker → the carrier's amplitude becomes correspondingly smaller.

**AM = Change the carrier's amplitude according to the analog information signal**

**AM Bandwidth:** If the original information signal has bandwidth B, the resulting AM signal requires a bandwidth of:

**BAM = 2B**

This resulting bandwidth is centred symmetrically around the chosen carrier frequency fc. This '2B' relationship reflects the fact that standard AM produces two symmetric sidebands (an upper and a lower sideband, each of width B) around the carrier.

**Figure 7.41** *Amplitude modulation*

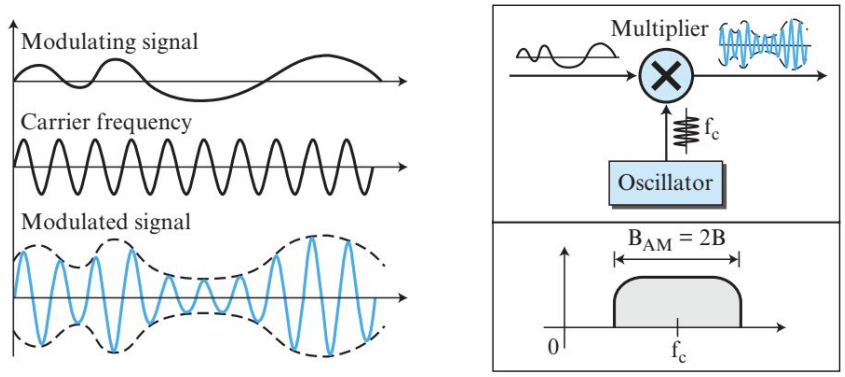



*Figure 7.41 — Amplitude modulation*

## 12.10 Frequency Modulation (FM)

In FM, the frequency of the carrier signal is varied in proportion to the instantaneous amplitude of the information signal, while the carrier's amplitude remains constant (the phase also changes, but only as an indirect mathematical consequence of the changing frequency, not as an independently controlled parameter).

**FM = Change the carrier's frequency according to the analog information signal**

FM signals are commonly generated in practice using a Voltage-Controlled Oscillator (VCO) — a circuit whose output frequency is directly controlled by an input voltage, which in this case is simply the information signal itself.

**FM Bandwidth:** An empirically derived approximate expression (Carson's rule) for FM bandwidth is:

**$B_{FM} = 2 \times (1 + \beta) \times B$**

where B is the bandwidth of the original analog signal, and β is a factor (the modulation index) related to the specific modulation technique/deviation used. Because of this expansion factor, FM signals typically require considerably more bandwidth than an equivalent AM signal, but this greater bandwidth 'spends' spectrum to buy much better

noise immunity — which is precisely why FM radio broadcasts generally sound noticeably clearer than AM radio broadcasts, especially in the presence of interference.

**Figure 7.42** *Frequency modulation*

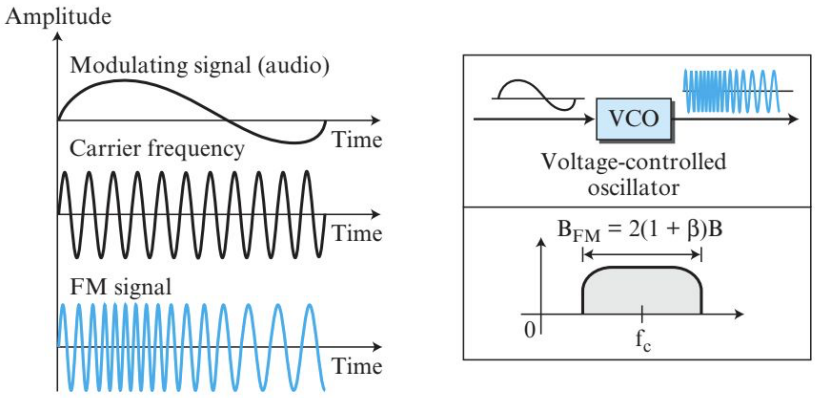



*Figure 7.42 — Frequency modulation*

## 12.11 Phase Modulation (PM)

In PM, the phase of the carrier signal is varied in proportion to the instantaneous amplitude of the information signal, while the carrier's amplitude and frequency remain constant.

**PM = Change the carrier's phase according to the analog information signal**

FM and PM are mathematically closely related to one another, but with one important distinction:

- FM: the instantaneous frequency deviation of the carrier depends directly on the amplitude of the information signal.
- PM: the instantaneous frequency deviation depends instead on the derivative (i.e., the rate of change) of the information signal.

PM generally requires a bandwidth several times greater than that of the original analog signal, similar in spirit to FM.

**Figure 7.43** *Phase modulation*

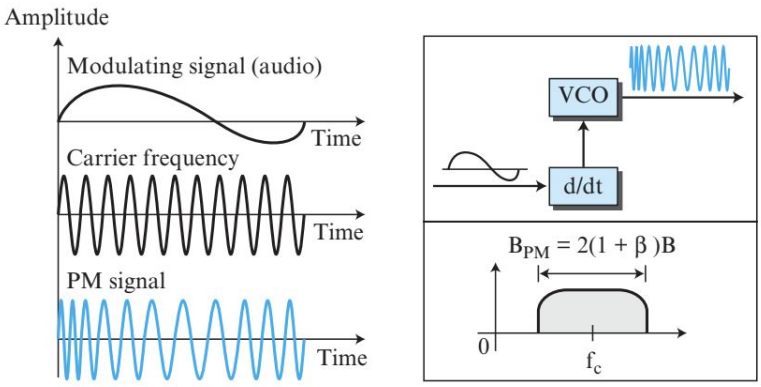



*Figure 7.43 — Phase modulation*

## 12.12 Summary: AM vs FM vs PM

| Technique | What changes? | Amplitude | Frequency |
|---|---|---|---|
| AM | Amplitude | Changes | Constant |
| FM | Frequency | Constant | Changes |
| PM | Phase | Constant | Constant (directly); varies indirectly with the derivative of the signal |

## 13. Bandwidth Utilization

Bandwidth is a scarce and valuable resource in any communication system. Bandwidth utilization refers to the set of techniques used to make the most efficient possible use of the available bandwidth on a link. Two major, largely independent techniques are used for this purpose, each with a different underlying goal:

| Technique | Primary Goal |
|---|---|
| Multiplexing | Combines multiple independent signals so they can share a single high-bandwidth physical link — the goal is efficiency. |
| Spread Spectrum | Deliberately increases (spreads) the bandwidth of a single signal — the goal is privacy and robustness against interference/jamming. |

### 13.1 Multiplexing — Overview

Multiplexing allows several independent signals, each originating from a different source, to share one single high-bandwidth communication link, rather than each requiring its own dedicated physical link.

**Many independent signals → Multiplexer (MUX) → One shared link → Demultiplexer (DEMUX) → Separate signals recovered**

A multiplexer (MUX) is placed at the sending end to combine the input signals, while a corresponding demultiplexer (DEMUX) is placed at the receiving end to separate the combined signal back into its original individual signals.

| Type | Main Idea | Signal Type Used With |
|---|---|---|
| FDM — Frequency-Division Multiplexing | Shares the available bandwidth by assigning each signal a different frequency band. | Analog signals |
| WDM — Wavelength-Division Multiplexing | Shares a single optical fiber by assigning each signal a different wavelength (colour) of light. | Optical signals |
| TDM — Time-Division Multiplexing | Shares the link by assigning each signal a different, non-overlapping time slot. | Digital signals |

Figure 7.44 *Dividing a link into channels*

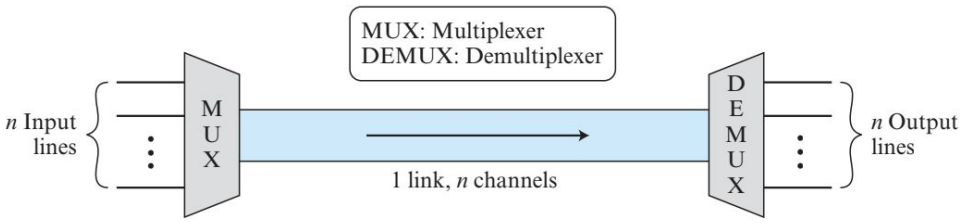



*Figure 7.44 — Dividing a link into channels (MUX/DEMUX)*

## 13.2 FDM — Frequency-Division Multiplexing

In FDM, the total available bandwidth of the link is divided into several separate, non-overlapping frequency bands, and each individual input signal is assigned/modulated onto its own distinct carrier frequency within this shared bandwidth.

**Worked Example:** Signal 1 might be modulated onto a 100 kHz carrier, Signal 2 onto a 200 kHz carrier, and Signal 3 onto a 300 kHz carrier — all transmitted simultaneously over the same physical medium, but occupying different, non-overlapping frequency ranges.

Small, deliberately unused frequency ranges called guard bands are inserted between adjacent channels to prevent one channel's signal from spilling over and interfering with its neighbours (a practical safety margin against imperfect filtering).

Figure 7.45 *Frequency-division multiplexing*

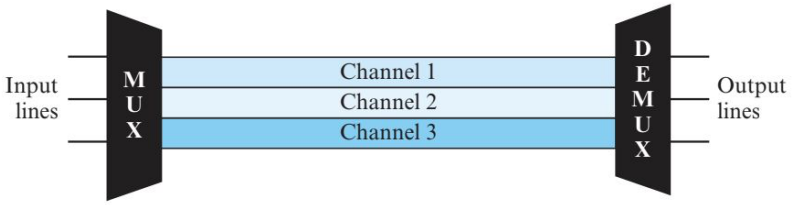



*Figure 7.45 — Frequency-division multiplexing (FDM)*

## 13.3 WDM — Wavelength-Division Multiplexing

WDM is conceptually identical to FDM, but is specifically applied to fiber-optic communication, where it is used to combine multiple independent light signals onto a single optical fiber, with each individual signal assigned a distinct wavelength (colour) of light.

**Many light signals (different wavelengths) → WDM MUX → Single optical fiber → WDM DEMUX → Separate signals**

WDM is the key technology that allows a single physical fiber-optic cable to simultaneously carry many independent high-speed data streams, dramatically increasing the effective capacity of fiber-optic backbone networks without needing to lay additional physical cables.

Figure 7.47 *Wavelength-division multiplexing*

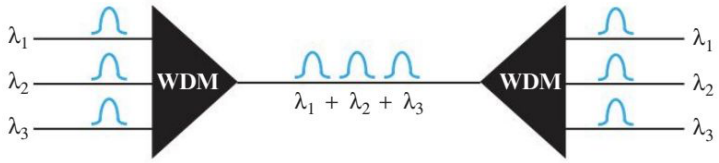



*Figure 7.47 — Wavelength-division multiplexing (WDM)*

## 13.4 TDM — Time-Division Multiplexing

In TDM, rather than dividing the available bandwidth by frequency, multiple users instead share the exact same link (and frequency range) by taking turns transmitting during different, non-overlapping time slots.

**User 1 → User 2 → User 3 → User 4 → User 1 → ... (repeating cycle)**

There are two principal types of TDM:

**Synchronous TDM:** Every input source is given a fixed, dedicated time slot in every round/frame, regardless of whether that source actually has any data ready to send at that moment. This is simple to implement, but can waste bandwidth if some sources are frequently idle.

**Statistical TDM (Asynchronous TDM):** Time slots are instead dynamically assigned only to those input sources that actually have data ready to send at a given moment, making much more efficient use of the available link bandwidth (at the cost of requiring additional addressing information to be sent, so the receiver knows which slot belongs to which source).

**Note:** Simple distinction to remember: Synchronous TDM = fixed slots for everyone (even if a source is idle, its slot goes to waste). Statistical TDM = slots are given out only when actually needed.

Figure 7.48 *TDM*

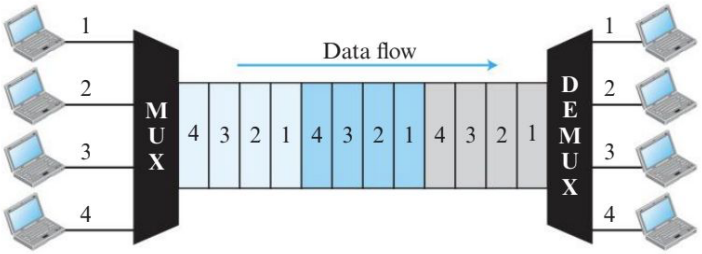



*Figure 7.48 — Time-division multiplexing (TDM)*

## 13.5 Spread Spectrum — Overview

Spread Spectrum (SS) is a fundamentally different approach: rather than trying to conserve bandwidth (as multiplexing does), it deliberately increases the bandwidth used by a single signal to be much greater than its original minimum requirement.

The purpose of doing this is:

- To improve privacy/security — making the transmission much harder for an unauthorized third party to intercept or decode.
- To reduce the effect of localized interference — since the signal energy is spread across a wide range of frequencies, interference at any single frequency affects only a small fraction of the overall signal.
- To provide anti-jamming capability — making deliberate jamming attempts far less effective, since an attacker would need to jam the entire spread bandwidth simultaneously.

If the original signal's minimum required bandwidth is B, spread spectrum techniques use a much larger bandwidth, BSS:

**BSS >> B**

There are two main spread spectrum techniques:

Figure 7.53 *Spread spectrum*

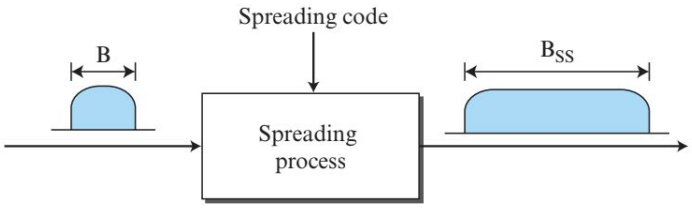



*Figure 7.53 — Spread spectrum*

## 13.6 FHSS — Frequency Hopping Spread Spectrum

In FHSS, the transmitted signal is not sent on one fixed carrier frequency. Instead, it rapidly 'hops' (jumps) from one carrier frequency to another, following a pre-agreed pseudorandom sequence known to both the sender and the receiver.

**Frequency 1 → Frequency 3 → Frequency 2 → Frequency 4 → ... (following the shared pseudorandom hopping sequence)**

Because the sender and receiver share the exact same pseudorandom hopping sequence (typically generated by a pseudorandom code generator seeded with a shared key), the receiver knows exactly which frequency to listen on at every instant, while an eavesdropper without knowledge of the sequence sees only brief, seemingly random bursts scattered across many frequencies.

Advantages:

- Very difficult for an unauthorized listener to follow or intercept the full transmission without knowing the hopping sequence.
- Reduces the practical effect of both narrowband interference and deliberate jamming, since any interference typically only affects a small number of the many frequencies used.

Figure 7.54 *Frequency hopping spread spectrum (FHSS)*

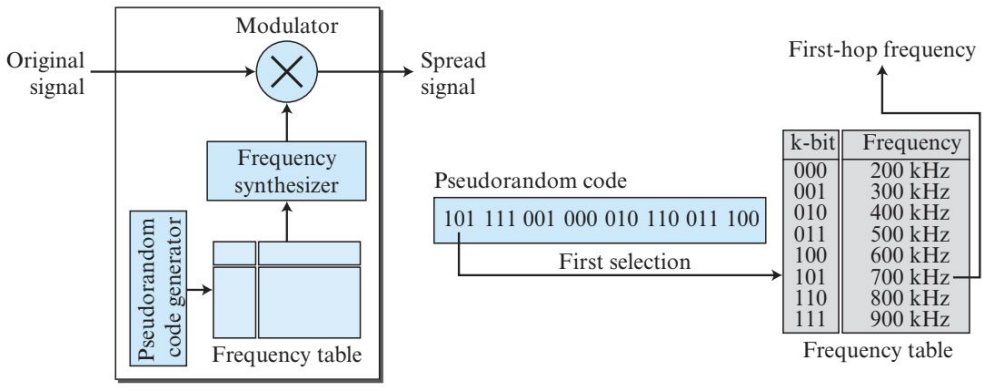



*Figure 7.54 — Frequency hopping spread spectrum (FHSS)*

## 13.7 DSSS — Direct Sequence Spread Spectrum

In DSSS, instead of hopping between different frequencies over time, each individual data bit is directly replaced by a longer sequence of smaller, faster bits called chips, according to a known spreading code.

**Worked Example:** Suppose each single data bit is replaced by 11 chips (using an 11-bit spreading code) — this is the classic '11-chip Barker sequence' historically used in early 802.11 Wi-Fi.

**1 data bit → 11 chips**

If the original data rate is N bits per second, then transmitting 11 chips for every original bit effectively multiplies the transmitted signal's rate (and hence the required bandwidth) by a factor of 11:

**Spread signal rate ≈ 11 × N**

Therefore, the required transmission bandwidth becomes approximately 11 times larger than that of the original, unspread signal — this is the deliberate bandwidth expansion that gives spread spectrum its name and its resilience properties.

Figure 7.57 *DSSS*

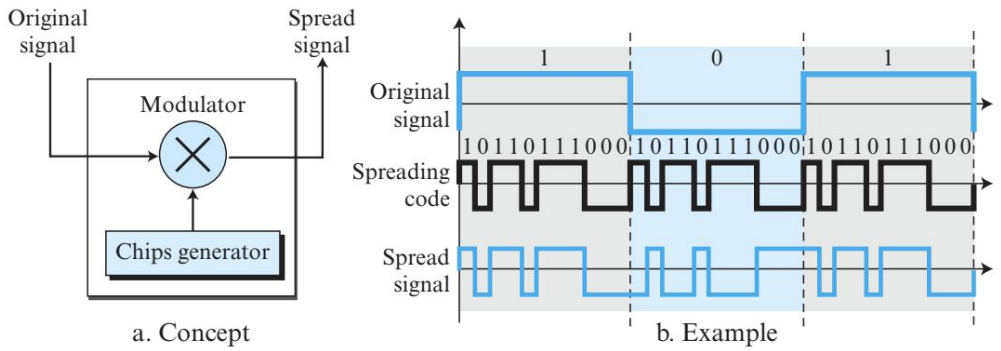



*Figure 7.57 — Direct sequence spread spectrum (DSSS)*

## 13.8 Multiplexing vs Spread Spectrum — Summary Comparison

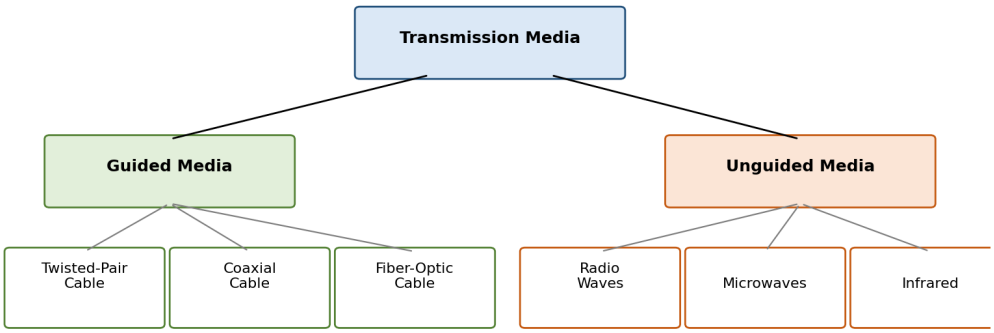
| Aspect | Multiplexing | Spread Spectrum |
|---|---|---|
| What it does | Combines multiple independent signals together. | Expands the bandwidth of a single signal. |
| Main goal | Efficiency (sharing limited bandwidth). | Privacy and anti-jamming robustness. |
| Bandwidth usage | Shares the total available bandwidth among users. | Deliberately uses much more bandwidth than strictly necessary. |
| Common techniques | FDM, WDM, TDM | FHSS, DSSS |

## 14. Transmission Media

A transmission medium is the physical path through which information actually travels from a sender to a receiver. In data communication, this medium may be a metallic cable, a fiber-optic cable, or simply free space (air/vacuum) through which wireless signals propagate. Because transmission media sit physically 'below' (outside) the Physical Layer's own protocols, yet are directly controlled and used by the Physical Layer, they are sometimes informally referred to as 'Layer 0' of the OSI model.

**Sender's Physical Layer → Transmission Medium (cable or air) → Receiver's Physical Layer**

Transmission media are broadly classified into two categories:




- Guided Media — provide a physical, bounded path (a cable) that signals are confined to travel along.
- Unguided Media — transmit signals through free space, with no physical confining boundary.

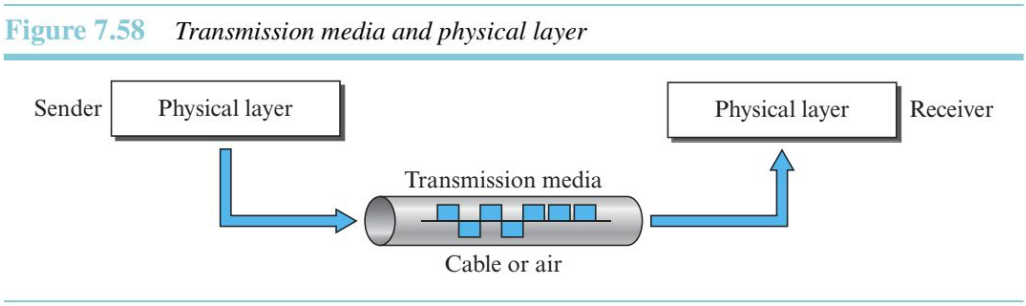



*Figure 7.58 — Transmission media and the physical layer*

### 14.1 Guided Media

Guided media physically confine and guide signals along a defined path. The three principal types used in networking are twisted-pair cable, coaxial cable, and fiber-optic cable.

### 14.1.1 Twisted-Pair Cable

A twisted-pair cable consists of two separately insulated copper wires that are twisted together in a regular helical pattern. This twisting is a deliberate design choice, not merely for convenience — it significantly helps to reduce noise and crosstalk (interference from adjacent wire pairs), because the twisting causes any external electromagnetic interference to be picked up nearly equally by both wires of the pair, allowing the receiver to cancel out this common-mode interference.

There are two common variants:

| Type | Description |
|---|---|
| UTP (Unshielded Twisted Pair) | The basic form, with no additional metallic shielding — lower cost, more flexible, but more susceptible to external interference. |
| STP (Shielded Twisted Pair) | Has an additional metallic foil or braid shield surrounding the twisted pairs, providing better protection from external interference, but making the cable more expensive, heavier, and less flexible. |

Twisted-pair cables are very commonly used in traditional telephone lines, DSL broadband connections, and Ethernet LANs (e.g. Cat5e/Cat6 cabling). A key limitation is that signal attenuation on twisted-pair cable increases as the signal frequency increases, which fundamentally limits both the achievable bandwidth and the maximum usable cable length.

**Figure 7.59** *Twisted-pair cable*

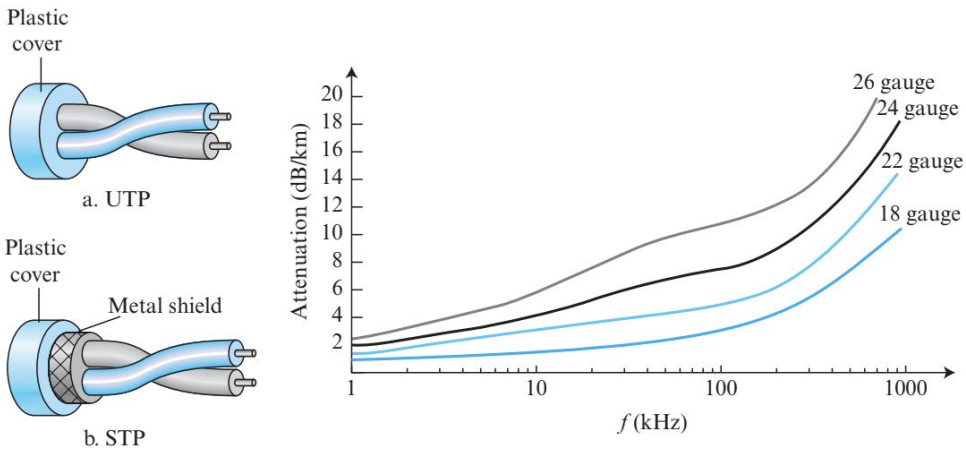



*Figure 7.59 — Twisted-pair cable (UTP and STP) and attenuation vs frequency*

### 14.1.2 Coaxial Cable

A coaxial ('coax') cable consists of a central copper conductor core, surrounded by an insulating layer, which is in turn surrounded by a cylindrical metallic shield (braid or foil), and finally an outer protective insulating jacket. This concentric ('co-axial') construction gives the cable its name. The outer metallic shield serves a dual purpose: it protects the inner signal-carrying conductor from external electromagnetic interference, and it also functions as the second conductor of the circuit (the return path).

Coaxial cable generally provides significantly higher bandwidth than twisted-pair cable and offers good inherent resistance to noise, thanks to its shielded construction. It has historically been widely used in telephone trunk networks, cable television (CATV) distribution networks, and early Ethernet LAN standards (10BASE2/10BASE5). In modern networks, however, coaxial cable in telephone backbone networks has been very largely replaced by fiber-optic cable, due to fiber's much greater bandwidth and lower attenuation.

**Figure 7.60** *Coaxial cable*

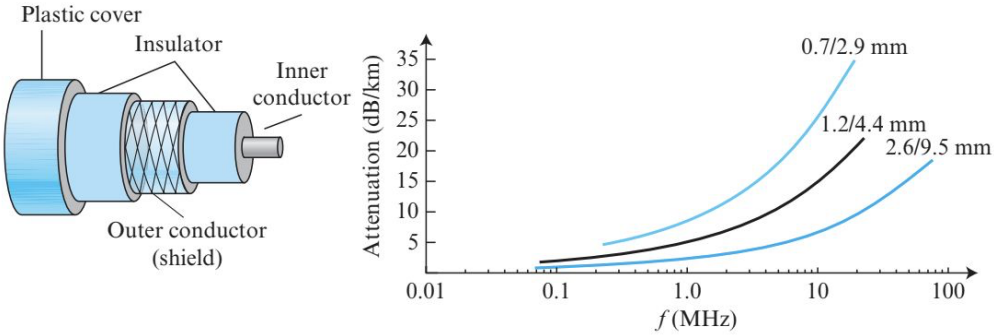



*Figure 7.60 — Coaxial cable and attenuation vs frequency*

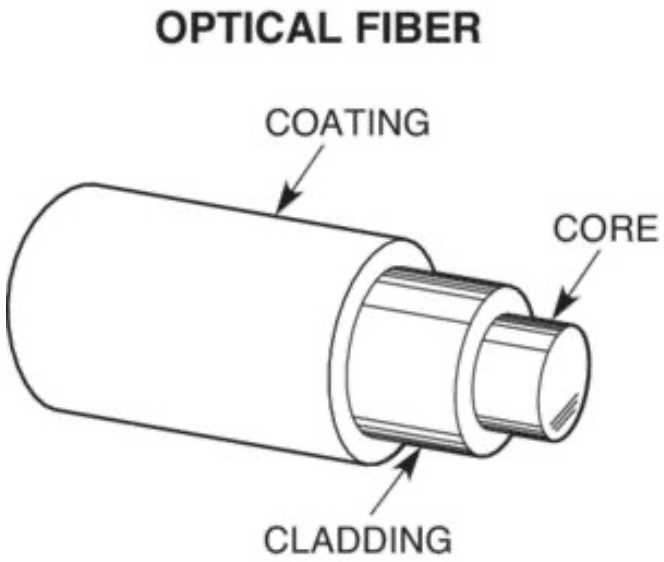
### 14.1.3 Fiber-Optic Cable

A fiber-optic cable transmits information encoded as pulses of light (rather than as electrical voltage/current) through an extremely thin strand of glass or plastic fiber. Light travels down the fiber core by a physical phenomenon called total internal reflection — the fiber's core has a higher refractive index than the surrounding 'cladding' layer, causing light to reflect internally back into the core whenever it strikes the core/cladding boundary at a shallow enough angle, effectively 'guiding' the light along the fiber even around gentle bends.

Fiber-optic cable offers several outstanding advantages:

- Very high bandwidth — capable of carrying enormous data rates, far exceeding copper-based media.
- Very low attenuation — signals can travel much longer distances before requiring amplification/regeneration.
- Excellent resistance to electromagnetic interference — since it carries light rather than electrical signals, it is essentially immune to electrical/RF noise and crosstalk.
- Enables true long-distance communication — used for continental and even trans-oceanic submarine communication links.

Fiber-optic cable is now the dominant technology for Internet backbone links, modern telephone trunk networks, cable TV distribution, data-center interconnects, and submarine (undersea) communication cables.




*Optical fiber structure: core, cladding, and coating*

## 14.2 Unguided Media

In unguided (wireless) media, signals travel through free space without being confined by any physical cable. The three major types used in networking are radio waves, microwaves, and infrared waves — these are distinguished primarily by their frequency range and, correspondingly, by how directional and how far-reaching they are.

### 14.2.1 Radio Waves

Radio waves are generally omnidirectional in nature, meaning they radiate outward in many directions simultaneously from the transmitting antenna, rather than being tightly focused in a single direction. Because of this, the transmitting and receiving antennas generally do not need to be precisely physically aligned with one another for communication to succeed.

Radio waves can travel long distances and — depending on their specific frequency — some can penetrate walls and other obstacles to a useful degree. They are very widely used for radio broadcasting and general-purpose wireless communication. However, radio transmissions are relatively susceptible to interference from other radio sources, and — because the usable radio spectrum is a shared, finite public resource — many specific frequency bands are tightly regulated by government/telecom authorities to prevent interference between different services.

### 14.2.2 Microwaves

Microwaves occupy a higher frequency range, approximately between 1 GHz and 300 GHz. Unlike radio waves, microwaves are generally unidirectional, meaning they must be tightly focused (typically via parabolic dish antennas) in a specific direction, and they require line-of-sight communication — the transmitting and receiving

antennas must have a clear, unobstructed path between them, with no significant obstacles (such as hills or large buildings) in between. Because of the Earth's curvature, long-distance terrestrial microwave links often require tall towers, or intermediate repeater stations placed at regular intervals, to maintain line-of-sight over long distances.

Microwaves are extensively used for cellular mobile communication, satellite communication, wireless LANs (Wi-Fi), and dedicated point-to-point backhaul links. Their inherently large available bandwidth allows for very high achievable data rates.

### 14.2.3 Infrared

Infrared waves occupy an even higher frequency range, from approximately 300 GHz up to 400 THz, and are primarily used for short-range communication only. A distinctive property of infrared radiation is that it cannot easily penetrate solid walls or other opaque obstacles.

This property is actually a double-edged sword: on one hand, it usefully prevents interference between infrared devices operating independently in adjacent rooms (since the walls effectively 'contain' each room's infrared signals); but on the other hand, this same property also fundamentally prevents infrared from being used for any long-distance or through-wall communication. Infrared is commonly used in TV/appliance remote controls and certain short-range indoor point-to-point data links.

## 14.3 Comparison Summary of Transmission Media

| Medium | Main Feature | Typical Examples |
|---|---|---|
| Twisted Pair | Low cost, widely available and used. | Telephone lines, Ethernet LANs |
| Coaxial | Higher bandwidth than twisted pair; good shielding. | Cable TV, older Ethernet standards |
| Fiber Optic | Very high speed and bandwidth; low attenuation. | Internet backbone links |
| Radio | Omnidirectional propagation. | Radio broadcasting, general wireless communication |
| Microwave | Directional; requires line-of-sight. | Cellular networks, satellite communication |
| Infrared | Short range only; cannot penetrate walls. | Remote controls |

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# MODULE 4 — QUESTION BANK

## A. 3-Mark Questions (Short Answer)

1. Define SNMP. Name its two main components (manager and agent) and state one function of each.
2. What is a MIB? How is it different from SMI?
3. List the symbols used in ASN.1 notation and give the meaning of any three.
4. Differentiate between analog data and digital data with one example each.
5. Define bandwidth of a signal. If a signal's frequency ranges from 1000 Hz to 5000 Hz, calculate its bandwidth.
6. State the Nyquist sampling theorem. What is the Nyquist rate for a voice signal with maximum frequency 4000 Hz?
7. Differentiate between bandwidth and throughput.
8. Define attenuation. Write the formula used to express signal loss/gain in decibels.
9. What is jitter? Why is it a concern for real-time applications?
10. Differentiate between baseband and broadband transmission.
11. What is line coding? Name any three line coding schemes.
12. Define quantization error. State the maximum possible quantization error for a given step size $\Delta$.
13. Differentiate between ASK, FSK and PSK based on what parameter of the carrier is changed in each.
14. What is a constellation diagram? What do the X-axis and Y-axis represent?
15. Differentiate between FDM and TDM.
16. What is spread spectrum? Name its two main types.
17. List any three types of guided transmission media.
18. Why is twisted-pair cable twisted? What problem does this solve?
19. Differentiate between UTP and STP.
20. State two advantages of fiber-optic cable over coaxial cable.

## B. 5-Mark Questions

1. Explain the working of SNMP with a neat diagram showing the manager, agent, and the direction of request/response and trap messages.
2. Explain the roles of SNMP, SMI, and MIB in network management, and how they work together to manage a device.

3. With examples, explain how new data types and subtypes are created in ASN.1. Illustrate SEQUENCE and SEQUENCE OF with suitable examples.
4. Explain analog and digital signals with the help of diagrams. List the three characteristics of a sine wave.
5. Explain the three causes of transmission impairment — attenuation, distortion, and noise — with suitable examples.
6. State and explain the Nyquist bit rate formula and the Shannon capacity formula. Under what conditions is each applicable?
7. Explain the steps involved in Pulse Code Modulation (PCM) with a block diagram.
8. Explain NRZ-L, NRZ-I, and Manchester line coding schemes with waveform diagrams for the bit sequence 0 1 0 0 1 1.
9. Explain the need for block coding. Describe the 4B/5B coding technique and explain how it solves the synchronization problem of NRZ-I.
10. Explain Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK), and Phase Shift Keying (PSK) with suitable diagrams, and state the bandwidth formula for each.
11. Explain Quadrature Amplitude Modulation (QAM) and state why it is more spectrally efficient than plain ASK or PSK.
12. Explain Amplitude Modulation (AM), Frequency Modulation (FM), and Phase Modulation (PM), and tabulate what changes and what remains constant in each.
13. Explain FDM and TDM with diagrams, and differentiate between synchronous and statistical TDM.
14. Explain FHSS and DSSS spread spectrum techniques with the help of examples.
15. Compare twisted-pair, coaxial, and fiber-optic cables based on bandwidth, cost, and noise immunity.
16. Explain radio waves, microwaves, and infrared as unguided transmission media, highlighting their directionality and typical applications.

## C. 7-Mark Questions

1. Explain in detail the architecture and working of SNMP, including the roles of the manager and agent, the three basic operations of SNMP management, and the relationship between SNMP, SMI, and MIB. Support your answer with a suitable diagram.
2. What is ASN.1? Explain why it is needed, and describe its built-in data types, the creation of new types and subtypes, and the SEQUENCE and SEQUENCE OF constructs, with suitable syntax examples for each.
3. Discuss data and signals at the physical layer. Explain analog and digital signals, the parameters of a sine wave (amplitude, frequency, phase), bandwidth, and the concept of composite signals via Fourier analysis, with appropriate diagrams.

4. Explain the three causes of transmission impairment (attenuation, distortion, noise) in detail. Derive/state the Nyquist and Shannon formulas for channel capacity, and explain the difference between them with a suitable example calculation for each.
5. Describe Pulse Code Modulation (PCM) in detail, covering sampling (and the Nyquist theorem), quantization (including quantization error), and encoding. Derive the PCM bit rate for a voice signal sampled at the standard rate with 8 bits per sample, and compare PCM with Delta Modulation.
6. Explain line coding, block coding, and scrambling as digital-to-digital conversion techniques. Describe polar, bipolar, and multilevel line-coding schemes, the 4B/5B and 8B/10B block-coding schemes, and the B8ZS and HDB3 scrambling techniques, with appropriate diagrams and examples.
7. Explain digital-to-analog conversion in detail, covering ASK, FSK, PSK, and QAM. Include the bandwidth formula for each technique, a discussion of the constellation diagram, and an explanation of why PSK and QAM are generally preferred over ASK in noisy channels.
8. Explain analog-to-analog conversion (AM, FM, PM) in detail, including the physical meaning of each technique, the relevant bandwidth formulas, and a comparison of what carrier property changes and remains constant in each. Explain why FM is more resistant to noise than AM.
9. Discuss bandwidth utilization techniques in detail. Explain FDM, WDM, and TDM (including synchronous vs statistical TDM) as multiplexing techniques, and FHSS and DSSS as spread spectrum techniques, clearly distinguishing the goals of multiplexing from those of spread spectrum.
10. Discuss guided and unguided transmission media in detail. Explain the construction, working principle, advantages, and applications of twisted-pair cable, coaxial cable, and fiber-optic cable (guided media), and of radio waves, microwaves, and infrared (unguided media), with a comparative summary table.